Flight Insights
Q22 design handoff

Network readiness. Automatic flight evidence.
Configuration-aware performance analysis.

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An integrated specification for Codex: cross-platform architecture, 45 diagnostic rules, one-click recording and collection, reliable parameter/BIN/TLOG association, and controlled network repair.

This is a design and research handoff. No application implementation, hardware validation or network change is claimed.

Read first

Sections 27-36 define the new automation and networking requirements. Sections 1-26 retain the original aircraft-health, PID/notch, energy and troubleshooting scope, with collection and baseline language updated. The accompanying JSON defines the implementation contract; CODEX_START_HERE.txt explains the build sequence.

Evidence rule

Make normal flights automatic. Make uncertainty visible. Never silently attach the wrong file or present missing evidence as a healthy aircraft.

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01 Decision And Scope

Build Q22 Flight Lab as an offline-first native desktop application for Windows, macOS and Linux. Use a Tauri 2 shell, React/TypeScript interface, and a packaged Python analysis engine. This is a design and research handoff for Codex, not an application release.

The product has two linked workflows: record and monitor available health signals during a flight; collect and deeply analyze the complete flight evidence after landing. The ground station remains responsible for flight control. The app explains measurements, identifies problems and proposes controlled investigations or tuning candidates. It does not replace ArduPilot failsafes or directly control the aircraft.

Make collection automatic after initial setup: Prepare & watch starts live capture, obtains parameter evidence, detects arming and creates the flight record; post-flight collection follows when disarmed. Manual recording/import/Collect now remain fallbacks. A post-flight action cannot recreate telemetry or radio history that was never recorded. Sections 27-36 specify the full automation, network diagnostics and association contract.

The preferred user experience is an installed app with its runtime bundled. Users should not need Python, Node, a terminal, an API key or a cloud account. Tauri supports packaged external binaries; platform-specific building and validation are required [R01]. A hosted-only website is a poor primary fit for unrestricted log sizes, unattended local collection and native device access.

The analysis core is deterministic and versioned. Optional language-model explanations can be added later, but measured values, fault classification and parameter proposals must come from inspectable algorithms and source evidence. An AI-generated narrative is never a substitute for a validated diagnostic rule.

This specification is intentionally broader than a simple graph viewer. A complete release includes data-quality checks, aircraft health, control/filter analysis, energy comparisons, collection recovery, issue tracking, immutable tuning baselines and explainable live alerts. Features awaiting actual device samples must be labeled unvalidated or unsupported, not represented by synthetic success.

Decision	Selection	Reason / boundary
App type	Local native desktop	Large logs, USB/SD access, offline field use and background recorders.
Initial platform	Windows x64 first; Linux x64 and macOS next within the same architecture	Matches the Mission Planner ground station; cross-platform installers remain a release requirement.
Main action	Prepare & watch	Automatic recording, flight detection and post-flight collection under an enrolled profile; manual collection remains available.
Tuning authority	Review and export proposals only	No automatic parameter writes, arming, mode changes or flight commands.
Data location	Local by default	No mandatory cloud upload, account or live internet dependency.

02 Q22 Context And Configuration Identity

Preserve the following project facts as profile inputs. Values reported in earlier work are not substitutes for current firmware readback or measurements.

Name the protected baselines “Q22 original - pre-FACt” and “Q22 with FACt tune”. Preserve FACt's original source file separately as provenance. Never manufacture the missing pre-FACt baseline from a guessed inverse diff. Resolve the exact accepted files before implementation seeds those labels.

Reported target firmware includes Cube 4.7.1-era work and a separate H7 custom build. Identify each flight from logged version/board information, AUTOPILOT_VERSION where available and the actual parameter snapshot. Pin matching metadata and source revision. A filename containing a version is a hint, not verified firmware identity.

Maintain a hardware-evidence register from the current Q22 documentation. Preserve later recorded closures of power/insulation/rail issues and confirmed Cube/Here4 yaw-180 mounting; do not resurrect old commissioning warnings automatically. Logs alone cannot prove physical motor direction or electrical assembly. Store current evidence and hardware revisions separately from flight diagnostics.

Area	Known context	Required handling
Primary aircraft	Cube Orange+; Airbot Mini Carrier PRO v2 / PDB Pro v2; ArduCopter; AUX2-AUX5 / SERVO10-13	Expected ESC log instances 9-12 from the existing review; confirm by individual motor identification on installed firmware.
Alternate aircraft	DEF Drones H7 Basic v3; Matek FCHUB-12S v2; separate supply/regulator arrangement	Separate profile and firmware metadata; expected output/ESC instances 0-3 require confirmation. Do not apply Cube assumptions to this build.
Propulsion	APD 120F3[X]v2; T-Motor MN605-S KV170; NS22x6.6 props; quad X configuration	Store motor/prop revision, rotation, physical corner, servo output, ESC instance and telemetry capabilities.
Energy	Reported 12S 40 Ah Xingto pack	Battery identity, chemistry, manufacturer limits, measured usable capacity, sensor calibration and temperature are required before limit/endurance advice.
Navigation	Here4; established installation/orientation	Capture device IDs, orientation, active sensor lanes, calibration and firmware. Same mounting does not justify copying sensor calibration offsets.
Ground/control	Mission Planner on Windows; QGroundControl may also be used; GX12 USB joystick	One controlling GCS. Separate received telemetry, operator inputs and actual onboard response.
Radio/video	DigiQ SkyPulse air and ground; SIYI A8 mini; 5.8 GHz RHCP/LHCP installation	Record actual radio/camera versions, antenna/cable changes, interfaces, codec and stream measurements. Do not assume a codec or undocumented metric.
Tune history	Keep original Q22 settings and the Q22 configuration incorporating FACt's tune	Maintain immutable named baselines and actual per-flight readback snapshots, plus independent candidate versions.

03 Product Requirements And Coverage Contract

“Monitor everything important” means every relevant subsystem is represented, with its evidence status and limits visible. It does not mean declaring a subsystem healthy when the necessary sensors or logs do not exist.

Each domain has five independent concepts: evidence availability, data quality, observed condition, diagnostic confidence and issue workflow state. Do not combine them into one aircraft-health percentage. A coverage indicator can describe the fraction of required evidence present, but its label must say evidence coverage and expose its denominator.

All findings must be reproducible from a source hash, parser/rule version, aircraft-profile snapshot and selected interval. The system must retain rejected or incomplete inputs in the collection manifest and explain why a metric or recommendation was withheld.

Domain	Required result	Examples of unavailable evidence
Data integrity	Identity, source coverage, gaps, corruption, duplicates, reboot boundaries, clocks, logging performance	A TLOG cannot replace high-rate onboard data or reconstruct downlink losses.
Vibration / IMU	Per-sensor vibration, clipping, gyro spectra, sensor errors and redundancy	VIBE is not a frequency spectrum; a post-filter trace cannot show unlogged pre-filter energy.
Control / tuning	Attitude/rate/position tracking, lag/overshoot, PID terms, saturation, filter trade-offs	Gain advice is withheld without representative excitation, noise evidence and actuator context.
Power / propulsion	Pack power/energy, voltage under load, ESC electrical/thermal/RPM data, motor balance and output demand	No individual-cell diagnosis without cell data; no thrust efficiency without thrust or a validated model.
Navigation / estimator	GPS quality, estimator innovations/status, lane changes, heading consistency and position/height response	A good GNSS fix alone cannot prove trustworthy navigation.
Control path / failsafes	Pilot command, GCS receipt/transmission, onboard input and response, mode/failsafe transitions	RCIN alone does not prove the GX12 USB input reached the GCS correctly.
RF / network / video	Both radios, HTTP collection health, link/throughput indicators, GCS network and video diagnostics	RSSI alone does not identify interference or explain a video freeze.
Controller / GCS system	Scheduler/load/logging health, resets, rails, disk space, PC sleep and recorder interruptions	Low-rate telemetry may miss a transient reset cause or short electrical event.
Configuration / experiments	Firmware-aware parameter diff, hardware revision, baseline regression, matched-condition comparisons	One changed flight does not prove a change caused an improvement.

04 Architecture And Technology Decisions

Use a modular desktop application, not a collection of separately launched scripts. Keep native access, numerical work and presentation behind explicit contracts.

The shell supervises one engine coordinator. The coordinator owns catalog writes and schedules bounded parser/analysis worker processes. Live recorders run independently of heavy FFT/import jobs, with queues and backpressure. A large post-flight import must not starve live recording. If the UI closes during recording, offer Keep recording in tray, Stop and close, or Cancel; never silently abandon a recording.

Use newline-safe framed IPC or JSON-RPC with explicit request IDs, protocol version, cancellation, progress and errors. Route diagnostic logging to stderr so it cannot corrupt protocol messages. Return references and bounded chart slices, not entire multi-gigabyte arrays over IPC. Handle sidecar crash/restart without losing completed archives or presenting stale values as live.

A local HTTP API can exist as a development-only alternative. If added in production later, bind loopback only and enforce per-launch authentication, strict origin/host checks and narrow endpoints. It should not be necessary for the first installed app.

Lock dependency versions only after the first cross-platform packaging spike. Do not copy exploratory environment versions from this conversation into a release. Verify Python wheels, native dependencies, webview behavior, serial access and filesystem permissions on clean Windows/macOS/Linux machines.

Electron is a reasonable fallback if Tauri's platform webviews block required interaction or packaging, but it adds a bundled browser runtime. Python/Qt is viable for a team prioritizing one implementation language; it is less aligned with the proposed rich web-style chart workspace. A PWA is useful as a future report viewer, not the main collector. The recommended choice remains Tauri plus Python.

Component	Decision	Responsibility
Desktop shell	Tauri 2 / Rust	Window, native file dialogs, allowed paths/devices, OS credentials, process lifecycle and packaging.
Interface	React + TypeScript + Vite	Stateful working surfaces, collection progress, findings, comparisons and accessible navigation.
Charts	Apache ECharts; narrow series payloads	Linked time-series, spectra, spectrograms, scatter plots and event overlays. Benchmark rather than assuming advertised point counts [R03].
Engine	Packaged Python with pymavlink, NumPy, SciPy and PyArrow	Decoding, units, segmentation, DSP, diagnostics, local collectors and report generation.
Metadata	SQLite, single writer	Profiles, sessions, manifests, alignment records, findings, experiment history and job state.
Numeric store	Partitioned Parquet; DuckDB queries	Out-of-core column selection and time-window queries. Parquet filter/projection pushdown supports this design [R02].

Component	Decision	Responsibility
IPC	Typed request/event envelopes over supervised stdio	Renderer invokes limited Rust commands; Rust routes to engine. No production HTTP server is required.
Packaging	Per-OS CI builds with bundled engine	Windows installer, macOS app/DMG, Linux AppImage and/or supported package. Sign/notarize release builds where required.

05 Internal Boundaries And Reproducibility

Separate collection, decoding, interpretation and advice so new log formats or firmware revisions cannot silently alter old conclusions.

The engine should also provide a headless CLI with the same analysis core for repeatable batch tests and engineering review. The desktop app remains the normal user entry point. CLI and UI must not have different numerical implementations.

Before reusing FilterReview/PIDReview or other third-party code, inventory licenses and notices for every file and submodule. ArduPilot WebTools identifies GPL-3.0 licensing; decide a compliant reuse/distribution approach rather than assuming it is permissively licensed. Pin commits and document intentional changes. A rewritten implementation still needs independent numerical validation [R07].

Use a content-addressed raw store (SHA-256) and immutable per-flight manifests. Stream to a temporary file, flush, hash and atomically commit; preserve file size and source metadata. Hash equality deduplicates bytes, not necessarily flight identity. Different source devices may legitimately contain overlapping observations.

Store data tables by flight/source/boot epoch/message group, with explicit schema versions. Do not load every message from a large raw-gyro log into Python dictionaries. Parse in batches, write Parquet, query only required columns/windows and compute plot decimation separately from metric calculation.

Never execute commands or render active markup from a filename, STATUSTEXT, radio response or imported note. Validate archive extraction paths, input sizes and decoder resource limits. Plugin adapters are trusted installed code; a log is data, not a plugin.

• Collector adapter: discover, probe capabilities, start/stop recording, list completed records, retrieve, resume and report a recovery action. Its output is original bytes plus a manifest; it does not invent engineering units.

• Parser adapter: sniff file structure, decode messages, retain source identity and raw timestamps, emit schema/unit metadata, records and corruption/truncation reports. Use file content as well as extension.

• Normalizer: map validated fields into typed signals, apply scaling exactly once, preserve instance/output mapping and mark missing/sentinel values explicitly.

• Clock service: create boot epochs, estimate or accept time transforms, retain anchors/residuals/uncertainty and refuse unjustified synchronization.

• Segmenter: propose armed/airborne/hover/cruise/climb/descent/turn/transient phases with confidence; let the user correct them. Segments have versioned definitions.

• Metric service: compute descriptive results using validated units and eligible intervals. Return denominator, data coverage, uncertainty and excluded intervals.

• Rule engine: evaluate versioned prerequisites, findings and hypotheses. Rules are deterministic, testable and explicitly distinguish published thresholds from engineering heuristics.

• Recommendation planner: convert supported findings into inspection, collection, configuration review or a bounded experiment. It must abstain from a parameter value when the evidence is insufficient.

• Report service: freeze the input manifests, versions, figures, findings and limitations into portable JSON plus human-readable HTML/PDF. Regeneration does not overwrite earlier reports.

06 Core Data Model

The machine-readable contract in this handoff defines the same entities. Add schemas and migrations before implementation relies on them.

Use integer microseconds/nanoseconds for persisted clock values. Preserve UTC epoch precision: serialize 64-bit absolute times as decimal strings over JavaScript-facing JSON, or use a typed binary format; use floating relative seconds only for plotting. Null means unavailable. Zero is reserved for a real decoded zero with valid quality.

Preserve original message definitions, FMT/FMTU units/multipliers and the parser's scaling behavior. Store both the original field reference and the normalized quantity. Unknown units remain unknown. A firmware upgrade can require a new mapping, not an automatic application of the previous scale.

SQLite is the catalog and issue history; Parquet holds large numeric arrays. DuckDB is a local query engine over those tables, not a second competing metadata authority. Limit concurrent writers and make migrations recoverable.

Entity	Required contents
AircraftProfileVersion	Aircraft ID; board/firmware identity; physical motor map; ESC instance mapping evidence; motors/props/battery; calibration status; hardware changes; limits and their provenance.
FlightRecord	UUID; aircraft profile version; operator/test notes; mass/payload/battery/weather; actual parameter snapshots; source IDs; proposed and confirmed flight intervals.
SourceManifest	Original name/location; SHA-256; byte size; device and software identity; acquisition method; start/end clocks; parser version; success/partial/error status; warnings.
SignalSchema	Source message/field; physical quantity; raw and engineering units; scaling provenance; sensor/ESC/system/component instance; expected and observed cadence; sentinels.
Sample / event	Source ID; boot epoch; source sequence/message offset; raw time; time domain; normalized time if valid; value or null; quality flags.
TimeAlignment	Source and target clocks; anchor pairs; affine/piecewise transform; offset/drift; fit residuals; uncertainty; confidence; user adjustments and provenance.
MetricResult	Algorithm/version; interval and phase; value/unit; count and time coverage; exclusions; uncertainty; source/field references; interpretation limits.
Finding	Rule/version; severity; evidence quality; confidence; observation; competing hypotheses; evidence intervals; next action; expected result; parameter applicability; retest criteria.

Entity	Required contents
ParameterSnapshot	Immutable captured/reconstructed values; scope, completeness, firmware and aircraft identity; acquisition/observation provenance. Reference-only imports remain unverified for flight use.
ParameterProposal	Separate unapplied candidate; baseline snapshot, proposed diff, reason, target firmware/aircraft, validation status and rollback reference. Never substitute for observed flight configuration.
Experiment / issue	Baseline and candidate; one functional change; confounders; planned success criteria; repetitions; results; open/investigating/retest/resolved history.
CollectionJob	Per-adapter state/progress; resumable transfer position; retries; chosen flight grouping; missing-data consequences and recovery instructions.

07 Collection Experience And Recovery

Configure once, then use Prepare & watch, optionally entered automatically on app launch. Arming creates an interval and disarming schedules collection. The following manual start/collect functions remain available for recovery; sections 27-33 govern the automatic workflow. Partial results remain useful.

• First use: select aircraft/profile; choose approved Mission Planner/QGC and camera folders; identify controller USB device; enter radio addresses; confirm optional forwarded telemetry. Each optional source may be skipped.

• Start flight recording: verify disk space, clock status and recorder reachability; open a session envelope; begin configured ground/RF/system recorders. Show live sample ages and whether each source is actually producing records.

• Collect flight logs: scan configured folders and mounted SD log directories; query a configured available controller after disarming; include RF/system captures; copy selected records to the raw archive; hash and validate each transfer.

• Match records using aircraft/system IDs, boot identity, time anchors and file intervals. File modification time alone is weak evidence. Propose ambiguous groupings and ask the user to choose; do not merge unrelated flights merely because they were imported together.

• Run completeness checks before deep analysis. Show each source as collected, partial, still recording, unavailable, unsupported, failed or not configured. State which analyses are blocked and provide a source-specific next action.

• Resume an interrupted transfer only after confirming device/log identity and consistency. Never mark a growing or incomplete remote log complete. Retain successfully collected sources if another adapter fails.

• Watch configured directories only after the user enables auto-collection. Detect stable file size/closed recordings before ingestion; treat open TLOGs as provisional. Avoid full-disk scanning, uncontrolled network sweeps or opening every serial port.

Failure	User-facing recovery
Controller port busy	Disconnect the GCS from that USB port, retry, or use its downloader/SD card. Preserve the controlling GCS during flight.
Aircraft armed / active log growing	Defer the onboard transfer until a completed log is available. Keep live recording active.
No TLOG found	Choose the actual GCS log directory; verify recording before the next flight. Do not claim recovery of packets never logged.
RF history absent	Explain that live capture was required; test the configured endpoints and start recording next time.
Unknown vendor format	Archive it, identify device/firmware and request one representative export/schema. Give manual export instructions when documented.
SD card or disk full / sleep interruption	Show the exact interrupted interval, preserve completed data and explain the recording gap.
No optional camera/system log	Keep flight analysis available; mark camera/system conclusions unassessed.

08 Source Adapters And Hardware Capabilities

Implement adapters as capability-driven integrations. “The model supports a feature” and “this installed firmware produced that measurement” are different facts.

For SkyPulse, the documented read-only GET endpoints are /scatter_data.json and /events.json, with about 50 seconds of history. Preserve every poll and failure. Dataset windows overlap; derive deduplicated samples only after validating the exact sample identity/time schema. Collection time bounds when the reply was observed, not when every point inside it occurred. A five-second poll interval is a proposed starting point to bench-test, not a guaranteed vendor requirement [R10].

Do not compete with Mission Planner/QGC for a serial connection or silently alter MAVLink stream rates. Maintain an app-owned recorder where a validated feed is configured, and preserve native MP/QGC TLOGs as complementary evidence. Use explicitly configured forwarding/router duplication; one-way forwarding does not provide a read-request return path. Record router configuration and traffic counters. Log download should be deferred while airborne and should not saturate the control radio link.

No automatic flight command, parameter write, log deletion or firmware update belongs in the collector API. Read-only parameter/version requests and log transfers must be explicit capabilities with tests that inspect all outbound MAVLink messages.

Adapter	Required behavior	Evidence boundary
ArduPilot DataFlash	BIN and textual DataFlash LOG; FMT/FMTU-aware schema; parameters, events and raw/batch sensor records	Decode supported firmware schemas; retain unknown message types and report undecoded fields.
GCS MAVLink	Mission Planner/QGC TLOG and documented forwarded stream; preserve system/component IDs and receive timestamps	Separate GCS-sent, vehicle-sent and routed/duplicate messages where the capture allows it.

Adapter	Required behavior	Evidence boundary
Controller download	Read-only LOG_REQUEST_LIST/DATA/END baseline; optional tested MAVFTP acceleration; SD import fallback	Capability-probe firmware and paths. Do not require newer virtual paths on a 4.7.1-era controller [R11, R12].
SkyPulse air/ground	Configured read-only statistics/events polling with raw responses, UTC and monotonic collection times	Manual endpoints expose rolling history. Firmware response structure and units must be verified from actual samples [R10].
APD/ESC	Consume available onboard/telemetry RPM, electrical, thermal and error fields; record unsupported channels	Plain BDShot RPM is not evidence of electrical/thermal coverage. Extended telemetry capability depends on ESC/firmware [R09].
Parameter exports	Mission Planner name/value and QGC formats; whole snapshots plus logged changes	Snapshot/readback identity matters; logged startup parameters may not establish an entire later flight if settings changed.
EdgeTX / GX12	Optional enabled SD Logs CSV import from the radio's mounted storage after flight	USB joystick operation does not guarantee aircraft telemetry is available on the radio. Do not change USB mode during flight [R31].
SIYI A8	Local media/sidecar import and documented Ethernet download where supported; camera metadata and events	Media retrieval does not imply durable diagnostic logs or measurable end-to-end latency [R19].
Ground computer	Optional local timestamped recorder for disk, CPU, memory, network and process/recorder health	OS-specific capability and permissions must be tested; no hidden claims of GPU decoder or joystick visibility.
Generic exports	CSV/JSONL mapping wizard with explicit units, clocks and instances	Never guess whether RSSI is dBm, a percentage or a vendor scale. Save the mapping with firmware/version.

09 Time Alignment And Flight Segmentation

A single pane of glass needs a shared timeline when evidence supports one, and visibly independent clocks when it does not.

Preserve these domains independently: FC boot/sample time; GPS week/time-of-week or validated UTC; TLOG ground receive time; PC wall clock and monotonic clock; radio-internal time; camera media time. Split at FC reboot, clock discontinuity, capture restart and counter wrap. Never sort two boot epochs into one apparent continuous flight.

Preferred alignment anchors are valid logged GNSS time paired with FC time, and MAVLink SYSTEM_TIME with valid Unix/boot values. Validate leap-second handling, time validity and source system/component. Estimate a piecewise affine transform from multiple anchors when drift matters. Store the chosen anchors, method, residual distribution and uncertainty [R13].

TLOG receive time includes link and scheduling delay. Shared arming/mode events are fallback anchors with bounded uncertainty, not proof of millisecond synchronization. Cross-correlation is a proposed alignment aid and must be labeled estimated; correlation is not causal attribution. Radio timestamps with undocumented epochs remain unaligned until verified.

Show alignment uncertainty as a range when users compare events across sources. Suppress claims that require ordering finer than the uncertainty. A statement such as “the RF drop preceded the rate error by 40 ms” is invalid if alignment uncertainty is 500 ms.

Detect armed intervals separately from airborne intervals. ARM or heartbeat state alone is not takeoff. Propose takeoff/landing from multiple available signals such as altitude change, vertical velocity, output demand and landed state. Uncertain phases must remain editable and visible.

Classify hover/cruise/climb/descent/turn/transient phases using recorded motion, commands and mode, with configurable thresholds and minimum dwell time. Initial engineering defaults can be proposed and tested, but are not universal ArduPilot limits. Store the segmentation algorithm/version and user corrections.

All metrics must state their interval and exclusions. Do not count disarmed bench time as flight energy, bridge gaps in integrated power, mix pre/post-parameter-change segments, or compare aggressive maneuvering against steady hover as an efficiency result.

10 Diagnostic Policy And Recommendation Levels

Every insight begins with an observation and can advance only as far as the evidence allows.

Severity and confidence are separate. A potentially serious but poorly evidenced condition can warrant investigation without claiming its cause. Use an evidence gap state for missing data, not a red aircraft-fault alarm. Published limits, hardware limits, user acceptance criteria and engineering heuristics must be labeled distinctly.

Each rule specifies eligible phases, minimum coverage/sample rate, debounce/dwell, hysteresis, prerequisites, exclusions, reference and firmware applicability. If a prerequisite fails, produce a specific “cannot assess” result and collection/test guidance. Avoid arbitrary global health scores.

Recommendation cards show the observation, source/fields/instances, time interval, confidence, possible causes, proposed action, expected benefit, possible trade-offs and a retest criterion. A parameter name listed as relevant is not itself a recommendation to change its value.

Do not infer physical RF packet loss from MAVLink sequence gaps alone, exact motor damage from asymmetric current alone, PID direction from high current alone, or a notch center from a VIBE summary. A diagnostic model must be validated against representative labeled fixtures before its conclusions are promoted beyond exploratory status.

Level	What the app may say	Required support
Observation	“IMU 1 clipping increased during this interval.”	Valid decoded samples, counter handling, exact source/time references.
Hypothesis	“Mechanical vibration is a plausible contributor.”	Multiple relevant observations and competing explanations; confidence stated.
Investigation	“Inspect the mounting, then capture pre/post gyro data.”	Action addresses an evidence gap or plausible cause; expected result defined.
Configuration review	“Check the selected notch tracking source against actual RPM coverage.”	Exact firmware metadata and actual source capability; no assumed bitmask.

Level	What the app may say	Required support
Parameter candidate	“Evaluate this proposed change set in this bounded test.”	Validated model/method, data sufficiency, constraints, before/after values, applicability and rollback.
Confirmed improvement	“Repeated comparable tests improved these measured outcomes.”	Predetermined metrics, matched conditions, repetitions and uncertainty; no unexplained regressions.

11 Vibration, Clipping And Imu Analysis

Prioritize mechanical and sensor health before gain optimization. Keep each IMU instance visible.

Decode the actual VIBE schema. Modern per-IMU records and older Clip0/Clip1/Clip2 layouts require different mappings. Compute axis distributions, p50/p95/p99, maxima and time spent above the published 30 and 60 m/s² guidance bands. Use time weighting for exposure and sample quantiles for distribution; label each accurately. Brief landing impacts must remain distinguishable from persistent airborne vibration [R04, R14].

Clipping values are counters: compute positive increments within each continuous boot/sensor epoch. A counter decrease is a reset/wrap/format event to resolve, not a negative clipping event. Do not attribute the initial counter value to a selected interval. Report count, rate, onset, duration and flight phase; show individual sensors before an optional aggregate.

Correlate vibration with motor RPM/output demand, speed, commanded angular motion and aircraft configuration only after alignment is established. Inspect repeated RPM orders, speed-related growth, loose mounting/cable contact, propeller balance/tracking and motor bearing candidates. These associations narrow inspection; they do not prove the mechanical cause.

Compare redundant IMUs using calibrated quantities, instances and health/error records. Check sensor disagreement, clipping, sampling discontinuities, temperature association, active primary/lane changes and unexplained estimator divergence. Do not average away a defective sensor or merge data from different sensor orientations.

If persistent clipping or major mechanical vibration exists, defer optimization proposals that could hide the symptom. The next step should prioritize inspection, sensor installation verification and a comparable confirming log. A notch filter is not a remedy for accelerometer saturation.

Acceptance: fixtures must cover high-amplitude brief impact, sustained airborne vibration, older/newer VIBE schemas, a reset inside the file, missing one IMU, unknown scale, clipped signal with absent VIBE, and a selected window beginning after an existing counter increment.

12 Gyro Spectra And Harmonic-Notch Analysis

Implement proper raw-gyro analysis, including raw and batch formats. Frequency analysis of the low-rate VIBE envelope is prohibited.

Support eligible GYR records and ISBH/ISBD batch data through firmware-aware decoders. Reconstruct batch sequence, instance, sensor type, scaling, sampling rate and sample time; detect missing/duplicated batches. Preserve pre/post-filter identity. Verify this against a pinned ArduPilot FilterReview implementation and known fixtures; do not merely infer format from a field name [R05, R07].

Within stationary-enough segments, remove the mean or documented trend, apply a specified window and compute Welch PSD with recorded sample rate, FFT length, overlap, resolution and units. Report amplitude spectra and power spectra distinctly; use 20 log10 for amplitude ratios and 10 log10 for power ratios. Validate Parseval/known-sine fixtures. Do not resample across large gaps or claim resolution beyond the data duration.

Expose per-axis/per-IMU PSD, time-frequency spectrograms and RPM-order overlays. Convert verified mechanical RPM to rotation frequency using RPM/60. Electrical RPM or a pole-pair-scaled field requires the correct telemetry conversion first. Never apply motor-pole scaling twice. Blade-pass orders require the actual blade count, not an assumption.

Compare pre/post-filter noise only for matching sensor, phase, sample provenance and scale. Calculate band power reduction and residual peaks. A lower plotted post-filter peak is not automatically better if filtering adds harmful control-band delay or the flight conditions differ.

Candidate notch planning checks source coverage and freshness, whether dominant peaks track RPM/throttle/FFT, relevant harmonics, width, attenuation, minimum-frequency behavior, phase delay and controller load. Prefer the smallest supported filter set that addresses repeatable noise. Model with the exact firmware filter equations and parameters, including a second notch bank and dynamic options where available [R06].

Relevant parameter families include INS_HNTCH_*, INS_HNTC2_*, INS_GYRO_FILTER, gyro/raw logging configuration and applicable ESC/RPM source settings. Resolve each name, range, bitmask and reboot requirement from the installed firmware. Do not copy wiki numeric masks when prose, versions or source definitions disagree.

A numeric candidate requires verified spectral/RPM data and firmware-correct simulation. Show predicted attenuation and delay as modeled results, then require a confirmation flight with measured residual spectra, control response, CPU/logging health and temperatures. If raw data or RPM is insufficient, recommend the exact missing capture rather than invent a center frequency.

13 Attitude, Pid And Actuator Diagnosis

Evaluate the control chain: pilot/navigation demand, desired attitude/rate, achieved motion, PID contributions and available actuator authority.

Use ATT, RATE and PIDR/PIDP/PIDY where supported, plus actual output/motor-limit evidence. Compute phase-specific error RMSE, absolute-error quantiles, bias, lag, rise/settling time, overshoot, residual oscillation bands and repeatability. Rate-only records support fewer conclusions than full PID/output/noise evidence [R08, R14].

Step-response metrics require a suitable isolated command change, adequate amplitude, initial/final steady behavior, known sampling and enough post-command time. Use reference normalization and defined rise/settling bands. Reject overlapping commands, saturated maneuvers, clipping, major estimator resets and data gaps. Fit or average responses only across comparable maneuvers.

Estimate lag with a documented method and uncertainty; cross-correlation of highly periodic or unexcited data can be ambiguous. Frequency-response/coherence analysis requires sufficient excitation and correct interpretation of closed-loop input/output signals. Do not report gain/phase margins from an invalid transfer estimate.

Separate undertracking due to command shaping, actuator saturation, insufficient thrust, filter delay, wind/disturbance, estimator issues and gain selection. Inspect I accumulation/recovery, P/D activity, feedforward, Dmod/SRate and applicable flags. A high D-term RMS can reflect sensor noise, commanded motion or gain behavior; it is not a standalone instruction to reduce D.

Map RCOU/output channels through SERVOx_FUNCTION and the verified motor order; protocol and configured endpoints matter. PWM-like numerical bounds are not universal across DShot/scaled outputs. Use firmware motor-limit/allocator flags when available, and label output-near-bound heuristics as such. Distinguish total thrust, attitude/yaw authority and individual output limitations.

Parameter families to review include the applicable ATC_RAT_RLL_*, ATC_RAT_PIT_* and ATC_RAT_YAW_* gains/filters/limits, ATC_ANG_*, ATC_ACC_R_MAX, ATC_ACC_P_MAX, ATC_ACC_Y_MAX, ATC_INPUT_TC and relevant motor parameters. Presence and naming are firmware-dependent. Do not issue a generic recipe to increase P, reduce D or remove filtering.

A tuning plan first resolves sensor/mechanical/estimator issues, then validates filtering, then uses an appropriate controlled tuning method or ArduPilot AutoTune workflow with actual vehicle constraints. Any app-generated numerical gain candidate must be supported by a validated system-identification/optimization method; otherwise offer a test plan and evidence links. Preserve the previous accepted parameter snapshot and change one functional group at a time.

14 Power, Battery And Endurance Optimization

Energy analysis must distinguish measured consumption, comparable performance and a causal explanation for improvement.

Calculate simultaneous electrical power $P(t)=V(t)\times I(t)$. Integrate signed power over valid continuous intervals: $E_{Wh} = \text{sum}[0.5(P_i+P_{i+1})\times dt]/3600$. Select a declared primary battery monitor; do not sum an aggregate monitor with the same pack's constituent monitors or combine BIN/TLOG duplicates. Cross-check logged cumulative consumption only after validating its units/resets.

Retain negative current as a measured value and flag calibration/sign/returned-energy interpretation. Never silently clamp it or fill missing values with zero. Define gap tolerance from actual cadence and expected rate; exclude long or invalid intervals and show usable coverage. Distinguish total selected-interval energy from energy in only the observed portion.

Report calibrated pack voltage/current, time-weighted mean and peak power, Wh/min, phase-specific power, voltage under comparable demand, thermal trends and observed usable capacity when a valid discharge test exists. Mean cell voltage from pack voltage divided by cell count is not individual cell voltage or balance.

Hover efficiency comparisons require near-steady height/motion and comparable mass, payload, battery state/temperature, props and wind. Cruise comparisons require defined valid distance/speed and matched segments; report Wh/km only with appropriate overlapping power/navigation coverage. GPS ground speed is not airspeed, especially in wind.

A speed-versus-power curve can identify an observed operating region for lower Wh/km or lower power within the tested envelope. It cannot establish a globally optimum cruise speed from a few unmatched points. Weight-normalized power is a comparison proxy with assumptions, not a direct propulsive-efficiency measurement.

Apparent voltage drop divided by current change can screen pack-plus-wiring resistance only over suitable short segments with near-constant state of charge and temperature; report fit quality and confounders. Do not label it intrinsic cell resistance. Compare pack and per-ESC voltage/current carefully for sensor calibration and timestamp differences.

Endurance estimates are scenarios based on measured/declared usable remaining energy, selected reserve and representative future power. Show assumptions and uncertainty; 40 Ah nominal capacity alone does not establish remaining energy or a safe return time. Do not infer battery chemistry limits from “12S”.

Connect tuning to energy only when oscillation/output activity and power co-occur under comparable conditions, and then frame a hypothesis. Validate by repeated matched flights with tracking/stability/thermal constraints. Lower PID gains can worsen control; lower filter noise can add delay. The objective is a trade-off among stable response, noise, thermal load, authority and energy, not minimum watts at any cost.

15 Escs, Motors, Airframe And Power Distribution

Use the exact physical-output-instance mapping and compare behavior within equivalent operating conditions.

For each configured motor, track available RPM, command, voltage, current, consumed charge, ESC/motor temperature and error/telemetry counters. Report channel completeness and freshness separately. All four RPM values do not establish four temperature/current streams. Check supported extended telemetry and the APD firmware before choosing a collection method [R09].

Analyze motor asymmetry only in stable, comparable phases with known mapping. Control allocation, yaw demands, wind, CG, payload and motor direction can produce legitimate differences. Persistent unexplained asymmetry should generate inspection hypotheses and a bench/mapping/calibration check, not an automatic diagnosis of a failed motor.

Look for abrupt command-to-RPM mismatch, temperature rise inconsistent with demand, telemetry loss, rail/current excursions and repeated output limiting. Require multiple signals and adequate telemetry rate before assigning mechanical/electrical causes. Identify the observed anomaly even when the cause remains unresolved.

Trend airframe/prop changes with vibration orders, phase-specific energy, tracking and durability notes. Logs alone cannot measure frame stiffness, drag coefficient or structural integrity. Add external measurement/test results as evidence rather than inferring them from a single spectrum or endurance number.

Controller, accessory and main battery rails are different. Record the actual regulator/power-module topology for the Cube and H7 profiles. POWR samples can screen logged rail behavior but cannot qualify capacitor ripple, startup transients, connector heating or the reported local capacitor ratings.

Manufacturer operating limits and user acceptance criteria are profile data with a source/version and units. No universal ESC temperature, motor-current or pack-voltage threshold should be hardcoded as safe for every installation.

16 Navigation, Heading, Altitude And Controller Health

Troubleshooting should connect observed trajectory/control errors to estimator and sensor evidence without assuming a single cause.

GPS/GPA review covers fix type, satellites, HDOP and available horizontal/vertical/speed accuracy, update gaps, velocity/position discontinuities and receiver changes. Evaluate airborne intervals and required flight modes. A low satellite count or high dilution is a context signal; a reported fix is not a guarantee against interference or spoofing.

EKF review covers lane-specific innovation test metrics, status flags, timeouts, resets, lane/primary changes and aiding changes, linked to GPS, magnetometer, barometer/range and vibration data. Verify version-specific XKF/NKF scales; these test metrics are not raw covariance. For applicable XKF4 quantities, 1 marks the innovation acceptance gate, not the Copter EKF failsafe threshold. Failsafe diagnosis requires the configured FS_EKF_THRESH and matching filtering, multiple-quantity/state/timer

logic; prefer explicit recorded activation [R33].

Heading review compares available magnetometers, estimated heading, yaw innovations and current-dependent changes after accounting for orientation and calibration. Association between current and field/heading may indicate interference but does not establish a cause. Do not transplant compass offsets/device IDs from another aircraft because mounting matches.

Altitude/position response uses desired versus estimated/measured motion with correctly decoded frames and sign conventions. Distinguish NED down from upward altitude. Inspect climb/descent command response, vertical oscillation, barometer/range inconsistencies, position-hold error and navigation-controller output limiting.

Controller health includes firmware/board identity, reset/watchdog/crash indications when logged, scheduler timing/load, loop delays, logging drops/buffer pressure, IMU errors, temperature and supply rails. Correlate load increases with high-rate logging, filtering and telemetry. A missing interval may be logger overload rather than a flight dynamics event.

Decode events/error-resolution semantics from the matching firmware. A historic ERR entry is not automatically an active fault; an error-clear code may close an incident. Preserve the event sequence and explicitly distinguish “no resolution captured” from “fault remains present now”.

17 Radio, Gcs, Joystick And Video Troubleshooting

Represent the complete observation path: operator input -> GCS/routing -> radio -> onboard receipt -> aircraft response; video uses a separate return path.

Preserve both radios' available RSSI/quality/rate/drop/temperature/Tx-power data and raw API responses. Units, averaging windows, counter behavior and firmware are adapter metadata. Compare directional traffic and RF metrics to application demand without summing overlapping history windows. A local radio API failure may be an HTTP/network failure rather than an RF outage.

For MAVLink, separate each system/component/channel, sequence stream, send/receive direction when known, routing duplicates and signing/version metadata. Report received-stream gaps and message cadence. Sequence discontinuities alone do not measure physical RF packet loss; logger behavior, routing and multiple senders can produce them.

Track pilot input freshness and mode/command requests where available, acknowledgements and actual onboard state. If a command appears on the ground but no onboard response is captured, inspect the missing chain and alignment uncertainty. Neither absence in a partial log nor a COMMAND_ACK by itself proves physical aircraft behavior.

For the GX12 USB path, preserve the EdgeTX model/version and GCS joystick mapping as configuration artifacts. Optional OS/HID diagnostics can observe device disconnects/input timing only if the integration actually exposes them. RCIN records show what ArduPilot received; they are not raw USB joystick measurements.

Video troubleshooting separates source camera health, RTP/transport/network behavior, decoder/render load and display delay. Record codec/resolution/received bitrate/frame/decode-drop data when provided by the actual GCS/player. Avoid launching a second decoder/capture that materially loads the flight computer without an explicit configuration choice.

A recorded video or ping measurement alone cannot establish glass-to-glass latency. Provide a controlled bench latency test workflow with a synchronized visual/time reference. Correlate freezes with radio/GCS evidence only at supported clock accuracy. SIYI media download can be supported separately from diagnostic log availability [R19].

Ground-system health should record process/app version, network adapter/link changes, disk availability, sleep/resume, recorder failures and available CPU/memory/decoder indicators. Alerts must identify when the observer itself is unhealthy so the user does not mistake a stalled dashboard for stable aircraft values.

18 Live Monitoring And Alert Lifecycle

Live monitoring is an additional observer. Its availability and resolution depend on forwarded/recorded telemetry, and it cannot see all high-rate onboard evidence.

• Show aircraft-connected state separately from stream freshness and recorder-active state. Use receipt age and source timestamp/sequence progression; receiving the same old value repeatedly is not necessarily fresh data.

• Define expected cadence per signal/source. Use configurable dwell, hysteresis and clear conditions for alerts. Suggested engineering defaults must be labeled and validated against representative flights, not presented as official thresholds.

• Use eligible-phase rules to avoid a normal disarmed or startup state generating repeated flight alarms. Data gaps, out-of-range values and device faults are distinct incidents.

• Display serious supported observations prominently, with the source age and next diagnostic action. Optional sounds/notifications are user-controlled. Do not replace or modify ArduPilot/GCS alarms and failsafes.

• Deduplicate related symptoms into an incident while preserving individual evidence. Record first/last occurrence, count, active/cleared state, acknowledgement and subsequent recurrence.

• Acknowledge means the user saw it; it does not mean resolved. Only a documented test/retest can close an engineering issue. Preserve investigator notes and the evidence for closure.

• After flight, recompute eligible rules against the more complete onboard data. Label revisions to a live hypothesis and explain why the post-flight conclusion changed.

• On low disk, engine crash, recorder backlog or clock jump, prioritize recording continuity and explicitly mark affected intervals. The last displayed measurement must show stale age, not remain green indefinitely.

19 Ux And Screen Requirements

Use a clean engineering workspace: light main area, restrained navy navigation, blue actions, 16 px body text and at least 14 px regular labels, with progressive disclosure of technical detail.

The persistent header shows aircraft, selected flight/session, live/post-flight mode and the main collection action. First launch permits immediate file import without configuring optional integrations. A synthetic demo can be included, but must remain explicitly synthetic in all screens, exports and comparisons.

Default charts show a few related channels with compatible units, not every field on one axis. Provide saved workspaces for vibration, control, power, navigation and link troubleshooting. Selected windows are visible and can be reset; every derived value carries its scope.

An issue follows Open -> Investigating -> Change planned -> Retest needed -> Resolved, with reopen and evidence history. Aircraft tuning recommendations remain evidence-gated proposals. Host-network issues can offer a reviewed Apply repair operation with validation and rollback, as defined in sections

28-29. User notes, hypotheses and attachments belong alongside the finding.

Parameter comparison must show changed/added/removed values, functional groups, units, bitmask labels, firmware compatibility and calibration/device-specific fields. Never silently apply a snapshot to the connected aircraft.

Support keyboard navigation, visible focus, non-color status labels, screen-reader chart summaries, reduced motion and 200% text enlargement. Test compact ground-station displays as well as large monitors. Empty, loading, offline, stale, corrupt, partial and unsupported states need explicit copy and actions.

Screen	Primary content	Essential behavior
Overview	Flight identity, measured summaries, highest-priority findings, timeline and source coverage	Show full-flight versus selected/partial scope. Missing values read Unavailable, not 0.
Signals	Searchable units/instances; linked cursors; zoom; events; spectra and spectrograms	Display source, clock/alignment confidence and sampling quality. Preserve gaps and decimate only for display.
Insights / issues	Observation, evidence, hypotheses, action, expected result and retest	Click evidence to open the exact chart interval. Filter by subsystem, severity, confidence, phase and workflow state.
Collect	Prepare & watch, automatic session/collection progress, source setup and targeted recovery; manual controls remain available	Useful partial completion; manual import always available; ambiguous flight matches require review.
Compare	Flight/phase metrics and conditions; parameter snapshots and proposed changes	Protected pre-FACt/FACt baselines, effective parameter timelines, configuration-to-metric links and matched-condition comparisons.
Aircraft	Versioned hardware, firmware, mappings, calibration, limits and test notes	Separate declared from measured/verified values. Profile edits do not rewrite historic sessions.

20 Parameter Proposals And Controlled Experiments

The app should help the user improve Q22 systematically while preserving a known baseline and documenting trade-offs.

Each candidate change references the actual source snapshot, target aircraft and exact firmware metadata. Resolve parameter availability, range, enum/bitmask meaning, dependencies and reboot requirements. Separate relevant parameters from proposed edits. Reject incompatible boards, unknown fields, mixed firmware assumptions or a stale pre-change snapshot.

Preserve original Q22, Q22-with-FACt and later accepted tunes as immutable versions. A candidate is a branch with a purpose, hypothesis and rollback reference. Same frame/motors/weight/orientation supports considering FACt's tuning as a starting point; it does not eliminate aircraft-specific validation or authorize copying calibration.

Before a test, record the functional group changed, expected metric effect and regression constraints. Capture actual all-up mass, payload/CG, battery ID/state/temperature, hardware/prop changes, wind,

flight phase/mode and operator notes. Prefer repeated comparable trials; show missing/confounded conditions rather than normalizing them away.

Compare distributions and time exposure, not only peaks. Report absolute and relative changes with sample/flight counts and uncertainty. Avoid treating thousands of correlated samples from one flight as thousands of independent experiments. Use flight-level or block-aware resampling when estimating confidence.

For endurance-focused work, jointly assess tracking, stability, authority, thermal behavior, noise and energy. A lower-power result that degrades response is not automatically an improvement. Keep the objective and acceptance limits visible.

Export a human-readable change plan and machine-readable proposed diff only after the user reviews it. Applying changes remains a separate deliberate workflow in the established GCS for the initial product. A later write feature would need a separate design and validation decision.

21 Implementation Sequence For Codex

Build vertical slices with real parsers and evidence-linked results. Do not substitute a polished mock dashboard for working data flows.

Suggested repository areas: desktop shell; UI; engine/collectors; engine/parsers; engine/normalization; engine/clocks; engine/metrics; engine/rules; engine/recommendations; schemas; fixtures; integration tests; docs. Keep DSP/analysis independent of React and the shell.

Record architecture decisions and deviations. If a format/capability cannot be validated without an actual sample, implement the archive/unknown/recovery behavior and identify the exact sample needed. Do not invent an endpoint, unit, device value or successful test.

Performance targets are proposed engineering acceptance targets, not measured claims: responsive UI during a 2 GiB import; cancellation acknowledged within two seconds; bounded memory rather than proportional dictionary growth; cached-window chart queries normally under 300 ms on an agreed reference laptop. Select the laptop/fixture benchmark before using these as release gates.

Package the Python runtime and native dependencies. Validate Windows USB/GCS interactions, macOS signing/notarization and architecture builds, and Linux serial permissions/webview dependencies. Updates must migrate data safely, preserve raw logs and allow recovery from interrupted installation. Offline use must work after installation.

The revision-2 stage sequence in section 35 and the JSON contract is authoritative. The following domain deliverables remain mandatory but can be implemented in that revised sequence.

Stage	Deliverable	Exit condition
0 - Foundation	Repository, contracts, dependency/license review, installer/sidecar spike, fixture plan	App/engine handshake and clean install on target OS; raw archive and migration design proven.
1 - Useful collection and review	BIN/LOG/TLOG/params/CSV/JSONL import; local/SD discovery; RF recorder; manifests; basic coverage/health/power/vibration	Identity/schema/units, per-source boot epochs and explicit scope validated before metrics. Real fixtures reproduce findings; missing/corrupt data never appears healthy.

Stage	Deliverable	Exit condition
2 - Full evidence workspace	Cross-source alignment, advanced phase selection, comprehensive health/event channels, issue workflow, comparisons	Correct instance/clock/unit handling and source-linked charts across representative Q22 bundles.
3 - Control/filter/performance	Raw and batch gyro, spectra, filter modeling, PID response and actuator context, matched-condition energy analysis	Parity fixtures versus pinned trusted tools; evidence-gated candidate proposals and regressions.
4 - Native integration and live health	Validated USB/optional MAVFTP, forwarded telemetry, live alert lifecycle, supported camera/GCS adapters	Device tests and failure recovery; no unauthorized flight/parameter messages; recorder isolation under load.
5 - Release qualification	Installers, accessibility, large-log performance, migration/recovery, reports and operational guide	All mandatory acceptance cases pass; unsupported device capabilities and outstanding limitations are explicit.

22 Acceptance And Validation Plan

Use unit/property tests for numerical invariants, real-format fixtures for parser parity, integration tests for collection and manual device tests for hardware behavior.

Use pinned ArduPilot tools/source and labeled reference datasets for expected outputs, with tolerances justified by decoding, windowing and numerical precision. Review discrepancies instead of fitting tests to the new implementation. Generated fixtures supplement real logs; they do not validate the user's device integration.

For each rule, include positive, negative and abstention cases. The rule is not complete until it refuses a conclusion when prerequisites are absent. Every exported result must retain the evidence and analysis version necessary to reproduce it.

A release is not complete while essential requested capabilities exist only as mock data. A documented hardware dependency is an explicit outstanding item, not a passed acceptance gate. The app may ship a clearly labeled earlier milestone, but must not present it as the finished specification.

Test family	Mandatory scenarios
Data identity	Duplicate bytes, duplicate filenames with different bytes, two vehicles in one TLOG, multiple flights in one file, wrong profile and unknown firmware.
Units / schema	Older/newer VIBE formats; fixed-point scale applied once; missing/sentinel values; corrupt/truncated records; unknown units; very large input.
Clocks	FC reboot, counter wrap, PC wall-clock jump, drift, invalid UTC, delayed TLOG delivery and radio time with unknown epoch.
Numerics	Known constant/ramp power; long gaps; negative current; partial windows; battery-monitor duplication; clipping counters starting nonzero/resetting.
DSP / PID	Known sine/noise, missing batch, irregular sampling, aliasing, pre/post provenance; valid/rejected steps; saturation and low-excitation cases.

Test family	Mandatory scenarios
Coverage	One missing ESC/RPM/temperature channel; stale values; unsupported sensor; partial log; enough data to plot but insufficient to recommend.
Collection	Busy port, armed aircraft, growing log, byte gap, disconnect/resume, changed remote log, bad radio JSON, failed poll, overlapping RF windows, disk full.
Safety of interactions	Inspect every outbound controller message; no arming, mode, mission, parameter writes or log deletion. Read-only source failure cannot stop the controlling GCS.
UX / persistence	Fresh install, offline mode, crash/restart, profile revision, immutable baselines, report reproduction, keyboard/200% zoom and clear synthetic data.
Live observer	Recorder overload, UI crash, stale telemetry, duplicate timestamps, app sleep/resume, alert clear/recur and post-flight revision of a hypothesis.
Hardware validation	Representative Cube and H7 logs; four mapped APD ESCs; both SkyPulse responses; actual GCS forwarding; SIYI capability probe.
Comparison	Matched trials improve known synthetic metrics; confounded payload/wind/time scope blocks causal improvement claims.

23 Open Evidence, Research Limits And Next Inputs

The architecture can be implemented without further product questions. Device-specific claims and numerical tuning logic need the following evidence before validation.

Research distinguishes verified source facts from proposed product/algorithm design. Primary documentation can differ from a particular firmware build. The implementation must prefer actual log definitions and matching source/parameter metadata, with versioned adapters, over an unqualified “current docs” assumption.

The supplied existing Flight Data Review Rev01 established the three-record BIN/TLOG/RF workflow, preservation requirements and the baseline diagnostic order. This handoff extends it into a product architecture. It does not validate aircraft hardware, certify the tune or claim that all requested measurements already exist.

No application implementation is part of this deliverable. Any exploratory code produced before the user redirected this task is excluded and is not an approved foundation for Codex. Start from this specification and validated dependencies/fixtures.

Needed input	Why it matters	Default until available
One complete same-flight BIN + TLOG + RF JSONL bundle	Establish actual fields, cadence, identities and clock anchors	Implement fixtures and explicit unvalidated adapter state; no claim of real-aircraft validation.
Exact firmware/board and full parameter readbacks for Cube and H7	Pin units, masks, message schemas and tuning applicability	Treat reported version labels as provisional.

Needed input	Why it matters	Default until available
Actual air and ground SkyPulse raw responses + firmware	Validate array structure, timestamps, units, rolling windows and API errors	Preserve raw responses; no invented normalized radio metrics.
Verified ESC output/instance/pole and telemetry capability mapping	Avoid double RPM scaling or masking absent ESC channels	Use declared profile mapping, clearly marked pending confirmation.
Accepted original and FACt-based Q22 snapshot files	Preserve the requested baseline pair precisely	Do not synthesize a baseline or overwrite existing files.
Battery datasheet/limits, calibration and usable-energy evidence	Support meaningful voltage, consumption and endurance interpretation	Display measurements and assumptions; withhold unsupported limit or remaining-time claims.
Ground-station versions, actual log folders and forwarding capability	Design one-button collection and live observation on the real PC	Manual import remains available.
Camera/player diagnostics and a bench latency sample if needed	Separate camera, network and decoding issues	Media import only where supported; latency remains unmeasured.
Reference laptop and flight test protocol	Set fair performance and optimization acceptance criteria	Keep proposed engineering targets labeled as unbenchmarked.

24 Firmware Field And Parameter Research Catalog

Representative Copter-4.7.1 fields are listed here to guide implementation. Availability still depends on the build and logging profile. Preserve exact FMT metadata and source revision rather than treating this table as a universal parser schema [R20-R29].

Do not reduce time-stamped PARM records to the final dictionary when analyzing an earlier maneuver. Retain a configuration timeline and split affected metrics when parameters changed.

Important verified Copter-4.7.1 source traps: BAT.CurrTot is written as consumed mAh. BAT.RemPct=255 and BAT.SH=0 represent unknown values; BAT.Temp=0 is ambiguous without temperature-support context. Raw BCL cell=0 is unavailable; otherwise cell volts=(raw-1)/1000. Raw PM.Load=800 corresponds to 80% load. Check whether the selected parser already applied a transformation before applying it again. XKF4 test outputs are not raw covariance. Modern PSC values use SI and NED/down conventions. CMD records a mission item; MISE indicates execution start, not completion [R21-R29].

The field catalog is a discovery checklist. Only implement a diagnostic after its actual schema, scale, sentinel values and relevant firmware logic have been verified against fixtures. Do not copy a string list into a parser and assume semantic correctness.

Subsystem	Representative messages / fields	Related context / parameters
Identity / state	VER, MSG, ARM.ArmState/Method/Forced, MODE.ModeNum/Rsn, EV.Id, SPOL	Board/build identity; arming checks; explicit enum tables.
Schema / logging	FMT, FMTU, UNIT, MULT; time-stamped PARM; DSF.Dp/Bytes/FM*	LOG_BACKEND_TYPE, LOG_BITMASK, LOG_DISARMED, LOG_FILE_RATEMAX, LOG_FILE_DSRMROT.
IMU / gyro	IMU.I/EG/EA/T/GH/AH/GHz/AHz; VIBE.IMU/VibeX/Y/Z/Clip; GYR/ACC.SampleUS; ISBH/ISBD	Sensor IDs/orientation; INS_USE*; raw/batch options; exact unit/scaling lineage.
Rates / PID	RATE.RDes/R/PDes/P/YDes/Y/ROut/POut/YOut/AOut/AOutSlew; PID*.P/I/D/FF/DFF/Dmod/SRate/Flags	ATC_RAT_* gains/filters/limits, ATC_ACC_*_MAX, command shaping and sampling configuration.
Actuators	RCOU.C1…; RCO2; MOTB.LiftMax/BatVolt/ThLimit/ThrAvMx/ThrOut/FailFlags	SERVOx_FUNCTION; FRAME_*; MOT_PWM_TYPE, thrust model/voltage/current limits.
Battery / cells	BAT.Inst/Volt/VoltR/Curr/CurrTot/EnrgTot/Temp/Res/RemPct/H/SH; BCL.Instance/V1…V12	BATT_* monitor/calibration/threshold/timer/source settings; physical versus aggregate monitors.
ESC	ESC.Instance/RPM/RawRPM/Volt/Curr/Temp/CTot/MotTemp/Err; ESCX; EDT2 when supported	SERVO_BLH_POLES, SERVO_BLH_BDMASK, SERVO_DSHOT_ESC and applicable serial telemetry configuration.
GPS	GPS.I/Status/GMS/GWk/NSats/HDop/Lat/Lng/Alt/Spd/GCrs/VZ; GPA.HAcc/VAcc/SAcc/YAcc/Delta/RTCM*	GPS source selection/corrections; backend/units; optional receiver interference indicators.
Estimator	XKF3 innovations; XKF4.C/SV/SP/SH/SM/FS/TS/SS/GPS/PI; XKFS sensor selections; XKT timing	EK3_LOG_LEVEL, EK3_SRC*, EK3_IMU_MASK, affinity; FS_EKF_* configuration and implementation.
Compass	MAG.I/MagX/Y/Z/OfsX/Y/Z/MOX/MOY/MOZ/Health; XKFS.MI	COMPASS_* IDs/orientation/calibration/motor compensation; active yaw source.
Position / mission	PSCN/PSCE/PSCD desired/target/actual position/velocity/acceleration; PSON/PSOE/PSOD/PSOT; CTUN; CMD; MISE	PSC_*, navigation limits, altitude frame and origin; mission upload versus execution events.
FC power / system	POWR.Vcc/VServo/Flags/AccFlags; MCU; PM.LR/NLon/NL/MaxT/Mem/Load/InE/ErC/ErrL; RTDT	Actual rail specification; SCHED_* configuration; loop rate and board features.
Control / failsafe	RCIN; RCI2.OMask/Flags; MAVC.SS/SC/TS/TC/Cmd/Res/WL; MAV; MODE; ERR; MSG; TLOG ACKs	FS_GCS_*, RC_FS_TIMEOUT, RC_OVERRIDE_TIME, applicable radio/battery/terrain failsafe options.

25 Logging Profiles And Evidence Acquisition

Provide normal-flight and temporary diagnostic capture plans, generated from the actual aircraft firmware and verified after recording.

The existing Q22 review contains a normal logging configuration and temporary Cube raw-gyro guidance. The app should import that accepted profile as a documented reference after exact firmware verification, not silently push the same numbers to both controllers.

For every proposed logging change, show the name, current value, intended fields, estimated storage/CPU/traffic cost, firmware support, reboot/readback requirements and verification step. More logging is not always better if it causes dropped data or controller overload. Remote DataFlash streaming is optional and bandwidth-tested, not enabled by default on the control link.

When data is missing, distinguish hardware does not measure it, transport does not expose it, logging disabled, stream rate insufficient, recorder failed, file incomplete and decoder unsupported. Each outcome needs a different recovery action.

Plan	Required evidence	Validation
Normal flight	Events/identity/parameters; IMU health/VIBE; battery; control and actuator state; GPS/EKF; ESC and relevant compass/system health; GCS TLOG; RF recorder	Check the resulting fields, instance count, cadence, gaps, storage and CPU/logging pressure. A configured bit is not proof of collected data.
Filter investigation	Raw/batch gyro with sample timing and known pre/post provenance; RPM and relevant filter parameters	Verify raw encoding, actual sample rate, record continuity and logger load before interpreting FFTs.
PID investigation	Representative controlled RATE/PID/actuator/attitude records; filter/noise context; exact parameter timeline	Verify adequate excitation, no invalid saturation/clipping/estimator interval, and enough pre/post response time.
Power investigation	Calibrated voltage/current, cumulative counters, battery identity/state/temperature, mass/payload, motion/phase, available per-ESC data	Cross-check sensors; use matched conditions and preserve gaps; add external electrical measurements where needed.
Navigation investigation	Selected GPS/compass/IMU/barometer/range sources; estimator lanes/innovations/status; position controller and mission/mode events	Preserve coordinate frames and active aiding configuration. Additional EKF/replay logs are a targeted capture choice.
Control-path / video investigation	GCS diagnostic and send/receive capture where supported; actual joystick events; both radios; camera/player/PC metrics	Confirm direction and clock lineage, observer overhead and available latency reference.

26 Troubleshooting Workspaces

Offer symptom-based entry points that open a useful set of signals and a short evidence-driven decision path.

Symptom	Inspect together	Next discriminating step
Vibration / shaking	VIBE/clipping, gyro orders, rates/PID, outputs/RPM, phase and speed	Separate mechanical noise from control oscillation; inspect hardware or collect missing raw evidence.
Short flight time / high current	Calibrated power, mass/payload, battery state, speed/climb, RPM/output activity and temperatures	Compare matched phases; test sensor calibration and change one supported factor.
Poor hover / altitude oscillation	Desired/actual vertical motion, clipping, baro/range, EKF, thrust limits and power	Identify estimate error versus actuator/control limitation before changing altitude gains.
Yaw oscillation / drift	Yaw demand/response, heading innovations, CW/CCW output groups, compass/current and limits	Check commanded yaw, allocation/CG, source selection and interference context.
Unexpected mode or failsafe	Mode reasons, active error events, configured timeouts, GCS/RC freshness and aircraft response	Reconstruct the event against the matching firmware; distinguish observed trigger from suspected cause.
Missing / hot / uneven ESC	Mapped instances, RPM, telemetry age, current/voltage/temperature and demand	Verify mapping/capability/wiring and compare under equivalent demand; do not diagnose a stopped motor from missing telemetry.
Video freeze / delay	Camera/player frames, network/decoder load, both radios, traffic demand and TLOG continuity	Separate video-only failure from common link or ground-computer failure; run a controlled latency test if required.
App shows gaps / strange values	Source identity, units/sentinels, clocks/reboots, parser warnings, DSF/PM and recorder state	Correct the observation pipeline before drawing aircraft conclusions.

27 One-Click Operating Contract

Revision 2 adds automatic session orchestration, a Network & devices workspace, controlled network repair and parameter-to-performance attribution. The app remains an offline-first desktop analysis companion. Keep its implementation in the separate flight-insights repository; do not mix app source into the Q22 aircraft documentation repository. Q22 is a configured aircraft profile, not a hardcoded limitation on other vehicles.

The main action is Prepare & watch. On first setup, the user enrolls the aircraft, selects the drone interface and GCS integration, authorizes local folders, configures radio/camera endpoints, and enables automatic capture and post-flight collection. Subsequent launches can enter this mode automatically according to the saved preference. Normal successful flights require no repeated file selection or recording button. Manual import, manual start/stop and Collect now remain available.

Prepare & watch begins useful capture before arming. It checks disk space, power/sleep state, network paths, live telemetry and source freshness; obtains current parameters and identity; and arms the app's watcher, not the aircraft. Start timestamps and failures are durably journaled immediately. Short-lived radio records must not wait for the aircraft to arm. Auto-start at OS login is optional and visible; show a persistent tray indicator while recording.

Every configured source has a required/optional policy for the chosen experiment. Ready for recording means the required evidence checks passed. It is not an airworthiness approval and does not block or permit aircraft arming. If the aircraft arms before readiness, start or continue capture, assign a provisional flight and report exactly what is missing. Do not discard that flight or interrupt the pilot to run setup.

Automatic actions under the saved profile include passive inspection, read-only identity/parameter requests when permitted, append-only recording, matching high-confidence evidence, and retrieving closed logs after disarming. Network changes are offered through a concrete repair preview. Applying a repair is one user action plus any OS elevation; it is not silently included in every Prepare & watch run. Recorded settings and reviewed proposals are separate from permissions to write aircraft parameters.

Provide three completion dimensions: capture lifecycle, file-association confidence and analysis coverage. A flight can be finished while awaiting BIN, and its basic telemetry report may already be useful. Do not collapse these into a green Completed label. Mandatory evidence gaps remain visible until resolved, explicitly waived with a reason, or marked unrecoverable.

28 Network And Stream Diagnostics

Diagnose the actual data path in layers: selected interface and addressing; route and source-address selection; neighbor/device reachability; service reachability; packet arrival; valid protocol decoding; receiving application consumption; freshness and quality. Success at one layer never proves the later layers. A radio web page opening does not establish working video, and an unanswered ping does not prove a device is unreachable.

Model endpoints with role, device/profile identity, expected interface, IP/prefix or hostname, protocol, local and remote ports, unicast/multicast mode, allowed sources, credentials reference, receiving process and capability evidence. Learn observed endpoints without silently replacing the approved profile. Port numbers are configurable. For the current Q22 profile, UDP 5600 video and UDP 14550 telemetry are established starting values, not universal constants.

For the user's two-network example, Internet Wi-Fi may use a DHCP address on 192.168.0.0/24 with gateway 192.168.0.1, while the dedicated wired drone interface uses 10.0.0.5/24 to reach SkyPulse 10.0.0.2. The address 192.168.0.1 may be a gateway rather than the computer's own address; inspect it. With both devices directly reachable on their respective interfaces, no bridging, NAT or forwarding between these networks is needed. Do not add a default gateway or DNS server to the isolated drone interface. Preserve the Internet default route and DNS.

If both subnets genuinely share one physical Ethernet segment, a supported secondary address may be appropriate. It does not connect physically separate networks. If a device is behind a real router, add only a destination-specific route using a verified next hop and verify the return path. Do not invent 10.0.0.2 as a gateway. Detect overlapping VPN routes, duplicate subnets on multiple interfaces, stale persistent routes, DHCP/APIPA addresses, duplicate-IP indications and an incorrectly selected source IP. A failed ARP duplicate probe is not proof an address is unused.

Network check	Evidence / result	Recovery
Interface and route	Adapter stable ID, link state, addresses/prefixes, DHCP, gateways, metrics, selected route/source address to each endpoint, VPN context and neighbor resolution.	Select the correct wired interface; repair its address or a proven route; preserve unrelated networks. A route table alone cannot establish device reachability.

Network check	Evidence / result	Recovery
Firewall and permissions	Effective rules when readable, active profile/zone, interface, receiving executable, direction, protocol and allowed source. Distinguish permission denied or policy-managed rules from a detected block.	Propose a narrowly scoped rule for the actual receiving program/port/interface/source. Never disable the firewall or open an unrestricted port range.
UDP socket ownership	Local bind address/port and owning process, competing GCS instances, unexpected interface binding and recent receive counters.	Resolve the intended owner through the configured integration. Do not kill the process, seize its socket or rely on SO_REUSEPORT/SO_REUSEADDR to duplicate unicast traffic.
Telemetry reception	Decode MAVLink frames/CRC, source system/component, signed status where available, autopilot identity, heartbeat age and per-source message cadence. Separate raw UDP arrival from valid aircraft telemetry.	Check radio destination, GCS binding, forwarding and router return path. Do not send flight-control messages as a connectivity test.
Video transport and decode	Packet/source/cadence counters; RTP sequence/reordering where applicable; codec detection or session description; fresh decoded frames; parser/decoder availability; consumer process and queue/drop metrics where exposed.	Distinguish wrong destination, no packets, wrong codec, port occupied, missing plugin, decoder failure and excessive buffering. A file/stream URL suffix is not codec proof.
Radio and camera services	Bounded authenticated/read-only requests to configured management endpoints, roles/versions and capability states. Test the actual radio-to-camera path when exposed.	Diagnose endpoint/authentication/proxy limitations. A SkyPulse video proxy does not prove transparent IP reachability from the PC to the airborne camera.

UDP has no TCP-style connection handshake. A configured allow rule, an open socket, or a probe with no reply does not prove incoming UDP works or fails. Verification requires expected live packets and, for usable video, fresh decoded frames. Source ports need not equal destination ports. Passive packet observation may need an additional driver or privilege; if unavailable, report the specific blind spot and use receiver counters/logs. Never diagnose Firewall blocked solely from silence. Inspect RTSP negotiation and the chosen RTP transport separately; RTSP TCP success does not establish its media path, and broad UDP ranges are not a default repair.

Inspect the stream without taking it away from Mission Planner or QGC. Preferred evidence is their receiver diagnostics or an already configured fan-out. A direct test on an occupied port is offered only on the ground with a clear handover. If video duplication is needed, provide an explicit, separately qualified local fan-out with bounded queues; do not assume multiple UDP receivers all see every packet. Do not transcode merely to observe quality. Expensive media capture/decoding is optional and lower priority than control traffic and telemetry recording.

True camera-to-display latency requires an independent timing experiment or validated source/display timestamps. Jitter-buffer setting, ping round-trip time and receive-to-decode delay are distinct measurements. Report them separately; a 60 ms jitter buffer is not proof of 60 ms end-to-end latency. Kernel UDP buffer-size is measured in bytes and is separate from jitterbuffer latency in milliseconds [R44]. Preserve video freeze/drop intervals and correlate them with radio/host evidence only within known clock uncertainty.

29 Cross-Platform Repair And Q22 Compatibility

Use a small signed native helper for privileged host-network operations. The UI and Python parsers remain unprivileged. The helper accepts typed operations and allowlisted arguments tied to a repair-plan ID; it is not a shell-command runner. A plan contains evidence, exact before/after values, affected interface/program/endpoints, required privilege, expected interruption, rollback operations, verification steps, expiry and applicability fingerprint.

Repair lifecycle: inspect -> classify -> prepare plan -> user applies -> recheck preconditions -> make changes -> verify drone and Internet paths -> commit or roll back -> record outcome. Before modifying, recheck the selected adapter, effective policy and recent disarmed state. If flight state is unknown, do not assume it is safe to interrupt the link. Defer disruptive work until grounded status is established. Read-only monitoring continues.

Snapshot only what the plan may change and give each owned rule/address/route a stable identifier. Apply idempotently. Rollback removes or restores this plan's changes, not the entire firewall or network configuration. If another administrator changes the same object, show a conflict rather than overwrite it. For reachability-changing host edits, use a locally supervised timed rollback that survives UI exit. Persistent changes and temporary test changes must be distinguishable. A failed rollback remains an explicit unresolved result with saved recovery steps.

Platform	Inspection / implementation	Repair boundary
Windows	Native IP Helper/NetTCPIP and NetSecurity APIs or structured PowerShell results; interface GUIDs, adapter/address/route state, UDP endpoints and effective firewall policy. Store/compare typed data, not localized command output.	New-NetIPAddress can disable DHCP on the target interface [R38]. Never add an alias to the shared DHCP Internet interface with a naive command. Prefer the dedicated drone NIC; support a DHCP-preserving mixed configuration only after OS-specific testing, otherwise provide guided setup. Preserve enterprise policy and third-party firewall unknowns.
Linux	Netlink plus NetworkManager D-Bus where active; inspect the actual manager and effective firewalld/nftables/UFW arrangement. Include route policy tables and interface/zone membership.	Change the active connection profile through its owner, not transient ip commands that the manager later overwrites. Use NetworkManager checkpoints where supported [R41]. On unmanaged/systemd-networkd or unsupported policies, permit inspection and guided steps until a tested writer exists. Do not run competing firewall managers.
macOS	SystemConfiguration network services/preferences, interface and route state; inspect available application-firewall and packet-filter evidence separately. Identify local-network privacy permission separately from routing/firewall failures.	Use a signed privileged helper and supported network-service changes with backup/rollback. Application firewall authorization is not equivalent to a Linux-style per-port rule. Local Network privacy must follow normal OS authorization and cannot be bypassed with elevation. Do not overwrite global PF rules; unsupported managed/VPN/security policy receives a guided fix [R40, R46].

Cross-platform support means tested capability reporting, not identical privileged features everywhere. The capability matrix must distinguish automatic repair, diagnosis only, needs permission, policy blocked

and unsupported. Release qualification must demonstrate basic inspection and recording on all declared OS targets; never show Fix automatically if the adapter has no validated apply/rollback implementation.

Carry these Q22 decisions forward as an enrolled profile, with current observations taking precedence over stale assumptions: ground-computer drone address 10.0.0.5/24; SkyPulse ground endpoint 10.0.0.2; configured stream destination 10.0.0.5; H.265 over UDP 5600; MAVLink UDP 14550; SIYI A8 Mini on the 10.0.0.x side. Do not invent the unconfirmed A8 host address or credentials. A2 Mini belongs to a different project.

The selected Windows firewall policy from the prior Q22 work allows the relevant UDP ports on wired interfaces, from remote 10.0.0.2, across profiles, without pinning the local address. Preserve that selected policy as a named profile. Observe source addresses before suggesting an adjustment: a router/proxy can alter them. Explain any narrower per-program proposal and preserve the working rule until the replacement is verified. Do not silently switch the whole Ethernet network to Private or delete unrelated prior rules.

Mission Planner's confirmed Q22 installation uses system-wide MinGW GStreamer after the bundled installation was removed, with H.265 AutoConnect and HUD video. Inspect the library path/version actually loaded by MP, process architecture, PATH resolution, plugin registry, RTP H.265 depayloader/parser and d3d11 decoder availability. A successful gst-launch in a different environment is not proof MP uses the same runtime. Earlier curl_url/plugin/decoder failures belong in the diagnostic catalog, not as assumed current faults.

Preserve the verified pipeline structure: udpsrc -> rtpjitterbuffer -> rtph265depay -> h265parse -> d3d11h265dec -> d3d11download -> bounded leaky queue -> videoconvert -> BGRA -> appsink. Keep the recorded 50/60 ms buffer choices as versioned historical configurations; do not select either as universally optimal. Provide separate Mission Planner XML and command-line exporters with syntax fixtures, including typed caps where used, buffer-size when configured, and XML escaping only where syntactically required. Parse/round-trip the XML rather than blindly replacing quotes or omitting properties. Use QGC's own video/runtime settings; do not force MP's plugin path or XML pipeline onto QGC [R39].

SkyPulse's current documentation notes that received MAVLink traffic can determine the destination instead of the configured fallback [R10]. Observe the actual GCS source interface and incoming stream destination. A diagnostic probe must not accidentally teach the radio a new destination. Management API paths and effects vary by firmware; the manual itself uses inconsistent get_config/get_config.sh names. Probe documented read-only variants carefully and pin the supported adapter. Configuration writes can reboot the radio, so any future radio-destination repair is a distinct grounded, explicitly reviewed operation with backup and manual recovery, not a silently reversible host fix.

30 Session Orchestrator And Live Capture

Create one durable CaptureSession when Prepare & watch starts, with a random UUID independent of wall time. It can contain several ArmedIntervals and FlightSegments. Give each armed interval an immutable UUID immediately, a monotonically allocated display number scoped to the enrolled aircraft, and recorded start/end clock evidence. An armed bench test remains an armed session; classify it as a flight only when airborne evidence supports that. The date label is useful presentation, not identity.

Separate the live watcher's state from each interval's collection state. The states below describe related activities, not one blocking global state machine. After a disarm, the watcher returns to ready/degraded monitoring while a separate per-interval CollectionJob settles, downloads or awaits SD evidence. Several prior intervals can be collecting/reconciling/awaiting evidence while the current interval records. Persist and recover each job independently; schedule shared-link bulk work only while freshly disarmed.

Use FlightRecord as the canonical analysis record for exactly one ArmedInterval, including bench/unknown classifications; it references its parent CaptureSession and zero or more airborne segments. The original term FlightSession is a deprecated design alias for FlightRecord, not a second grouping. ArtifactAssociation and RetrievalRequest reference the specific FlightRecord/ArmedInterval as well as the CaptureSession. Optional sortie groups may link several records but never erase interval identity.

State	Transition / action	Failure behavior
DISCOVERING	Load the approved profile, identify the aircraft/GCS/endpoints, open the recorder journal and begin reachable short-lived-source recording.	Stay inspectable/offline with specific recovery; never bind the first unknown aircraft automatically.
PREPARING	Inspect network, establish the approved telemetry path, fetch identity and full parameters, check logging capabilities and disk budget; baseline available remote log inventory while disarmed.	Report missing requirements while recording whatever is available. A complete parameter cache from yesterday is not a fresh readback.
WATCHING_READY or WATCHING_DEGRADED	Continuously record with per-source freshness; observe arming and parameter changes; revalidate readiness after reboot, link gap, GCS/profile/interface change or updated parameters.	Degrade individual evidence states without stopping other collectors. Readiness is not permission to fly.
ARMED_RECORDING	On fresh autopilot heartbeat armed transition, attach the best parameter evidence and snapshot interval, assign armed-interval UUID/display number, record arming evidence and continue sources. Mark airborne segments separately.	If the app starts already armed, mark unknown start/preflight evidence and capture from now. Never create a fabricated preflight snapshot by reading parameters in flight.
LINK_UNCERTAIN	Missing heartbeats, connection loss or stale state preserves the open interval and last known state; record the gap and retry passive connection within the configured policy.	Silence is not disarming. No bulk download or disruptive repair while armed state is stale/unknown. Reboot detection splits boot epochs without assuming a normal landing.
DISARM_SETTLING	A fresh, valid disarmed indication closes the armed interval provisionally; retain a configurable post-roll and confirm logger closure/availability. Record subsequent rearming as a new armed interval.	Do not merge distinct arm cycles to simplify filenames. One physical sortie may group intervals only with explicit evidence/review; no takeoff means bench/test.
COLLECTING	Fetch postflight parameters and log inventory; select candidate onboard records; collect stable GCS logs and background artifacts; estimate transfer time; download only while the known aircraft is freshly disarmed.	Pause bulk transfers promptly if arming occurs or state becomes unknown. Preserve checkpoints; do not saturate the control link. Continue the watcher for the next flight.
RECONCILING	Parse identity, parameter timelines, boot/arming intervals and source overlaps; validate checksums/structure/completeness; assign evidence with recorded reasons.	Contradictory or ambiguous evidence stays quarantined from the confirmed flight bundle; offer a narrow recovery task.

State	Transition / action	Failure behavior
COMPLETE or AWAITING_EVIDENCE	Freeze a manifest revision and run eligible analyses. Late files create a new manifest/report revision while preserving prior provenance.	The readiness of the next flight is independent of unfinished collection for the previous one. Permanent missing evidence remains visible.

Use fresh autopilot HEARTBEAT state and correct source identity, not any component heartbeat or replay stream. Detect reconnect/boot boundaries using available boot counters/timestamps and identity; never use MAVLink packet sequence alone as a boot ID. Ignore replay-fed arming transitions for live orchestration. Maintain per-vehicle state and explicitly block enrollment ambiguity when two aircraft reuse a system ID.

The recording supervisor runs independently of chart/FFT jobs. Persist append-only chunks plus monotonic/UTC receipt times, raw bytes and offsets; checkpoint frequently enough to bound crash loss. Recover unfinished sessions after restart. Disk admission uses measured write rates, required-source budget and reserve. Low storage first stops optional media/verbose capture, alerts visibly and preserves essential small records where possible; never delete raw flight files automatically. PC sleep or collector crashes create explicit coverage gaps.

Record: app-owned timestamped MAVLink stream when a validated feed exists; GCS log references; raw SkyPulse statistics/events from both reachable units; network/route/socket changes; relevant receiver diagnostics; host CPU/memory/disk and collector health; parameter observations and configuration snapshots. Capture optional compressed video only under a selected storage policy. Lack of an air-unit management path must not hide the ground-unit recording or claim both sides were captured.

Radio polling starts before arming and continues through post-roll. Five seconds is an initial testable poll interval for the documented roughly 50-second rolling history, not a guaranteed device limit. Persist raw replies and transport failures before normalization; deduplicate overlapping history only after validating timestamps/identity. Backoff and jitter must remain visible, with an irrecoverable-gap event when retention may have been exceeded. A live source without a durable export is a required recorder adapter when its metrics are needed for the experiment.

31 Parameter Snapshots And Configuration Timeline

Associate measurements with what the aircraft reported, not the parameter file most recently opened on the computer. Each snapshot records target aircraft identity, system/component, firmware/build, board, boot epoch, transport, start/end monotonic and UTC times, names/types/raw values, expected/received counts, parser version, completeness, consistency observations and raw/canonical hashes. Save a human-readable .param export plus the richer manifest; .param alone cannot carry the evidence contract.

Read parameters during preflight while disarmed. Prefer a supported read-only packed-parameter MAVFTP download, with standard PARAM_REQUEST_LIST / targeted retry fallback. Capability-probe the installed firmware. Count, decode and validate types; accept and deduplicate identical retransmissions, but reject conflicting duplicate values/index mappings or unresolved missing entries. Treat each MAVLink component separately; complete autopilot parameters does not mean complete camera or every component. Monitor count/schema changes and restart affected downloads, with bounded retry and backoff [R34, R35].

For the packed adapter, validate the format magic and entry count, distinguish its internal type codes from MAV_PARAM_TYPE, and read through protocol EOF rather than trusting ArduPilot's estimated open-file length. Implement the documented fixed read-block/padding semantics with fixtures. For the

message adapter, validate count/index coverage by component and resolve names/types using the installed firmware's encoding. A GCS cache may substitute for another transfer only when its completeness, capture interval and identity evidence are exposed. Serialize large readbacks against other bulk transactions; avoid an endless refresh loop caused by continuously learned values.

MAVLink readback is not atomic. Packed FTP fixes the parameter set at open but reads values as blocks are generated; individual PARAM_VALUE synchronization can also miss updates [R34, R35]. A full, count-complete download therefore proves collection completeness over an interval, not an instantaneous configuration exactly at arming. Represent complete, consistent over observed interval, fresh, and corroborated at flight time as distinct properties.

A configurable initial freshness budget can be 60 seconds before arming, explicitly an engineering policy to validate. Refresh after known edits, firmware/profile change, reboot or coverage gap regardless of age. During the preflight quiet period, detect conflicting duplicate values or observed edits; retry the affected/full snapshot. A second matching read increases confidence but is not an atomic guarantee. Never block recording or claim the aircraft was prevented from arming when the snapshot is incomplete.

At arming, attach the newest eligible preflight snapshot and its age/coverage. Preserve all incoming parameter observations with timestamps during the flight; use observed changes as positive evidence only, not proof every change was broadcast. Avoid a bulk parameter read while armed. After disarming, fetch a full readback before the next configuration change where possible. If pre/post differ without a reliable change event, create an unknown-change interval; do not assign the postflight value retroactively to the whole flight.

Reconcile decoded BIN startup PARM and later PARM observations, captured MAVLink observations and pre/post readbacks under firmware-specific semantics. Startup dumping itself takes time; reading a PARM record does not by itself establish when a value became effective. Some settings take effect only after reboot or subsystem restart. Store observed_at and effective_from separately, with unknown boundaries allowed. Do not imply that a new parameter value immediately changed the active controller.

Build ParameterTimeline segments with an immutable snapshot/observation lineage, effective-time uncertainty and diagnostic eligibility. If AutoTune/learning or a user change alters relevant gains, split the analysis at the supported boundary and exclude uncertain transition windows. If the BIN supplies enough configuration evidence when the preflight read is missing, permit the supported analyses with a provenance label; do not retroactively claim a full preflight snapshot existed.

ParameterCapture is an acquisition-generation record whose output snapshot_id references an immutable ParameterSnapshot, complete or partial for its declared scope. ParameterObservation carries aircraft/component/boot identity and a source pointer; capture_id may be null for independent BIN observations. ParameterProposal is a separate unapplied candidate record referencing a baseline snapshot. EffectiveConfigurationInterval.base_snapshot_id can reference captured/reconstructed evidence only, never an unapplied proposal. Imported reference files remain reference evidence until verified against an actual flight.

Use a content hash of the exact original export/bytes, a canonical typed-value hash, and versioned semantic fingerprints by functional group. Canonicalization is explicit about sorting, component ID, parameter name/type and lossless numeric representation; never hash rounded display text. Include scope/completeness in the snapshot identity so a subset cannot masquerade as a full set. Types absent from a text export remain unknown or explicitly inferred, not falsely observed. Firmware/build and hardware-profile identity are separate required comparison keys. A changed raw file order can leave the semantic fingerprint unchanged.

Keep all values, including learned/calibration/device fields, in the authoritative snapshot. Default comparisons group PID/control, filters, motor/output, navigation, battery/failsafe, logging and device-specific/learned state. Joystick calibration, learned hover and device state may be collapsed by

default per user preference, but their presence and exclusions remain visible. Learned hover, mass-related adaptation and calibration still matter to relevant performance models. Never delete them to make two flights appear equivalent.

Rename the protected display baselines to Q22 original - pre-FACt and Q22 with FACt tune, following the later project decision. Preserve original source filenames/bytes and aliases for older Brian references. Never synthesize a missing pre-FACt baseline. Parameter proposals remain reviewed exports for the GCS; host-network repair permission does not grant aircraft-parameter write permission.

32 Automatic Bin, Tlog And Flight Association

Use a many-to-many evidence model. One BIN may span several armed intervals; a flight may span multiple source files or boot epochs; a TLOG may contain multiple vehicles and flights. Store the complete original once and attach vehicle/boot/time/byte-range selections to sessions. Do not trim or rename away its provenance. Byte-hash equality proves duplicate content, not the same flight.

Association evidence has independent dimensions: enrolled hardware UID when available; board/firmware and parameter fingerprint; endpoint/GCS provenance; boot-time mapping; armed/event sequence; overlapping movement/altitude/voltage patterns; validated UTC; onboard log inventory and transfer identity. AUTOPILOT_VERSION UID fields can be absent; system ID and VER firmware/board information are not unique physical identity. Missing UID lowers evidence, rather than causing fallback to filename as proof.

Use deterministic matching rules with hard contradiction checks before confidence ranking. Strong UID mismatch, incompatible boot/event sequence or non-overlapping supported intervals rejects automatic assignment. Prefer a direct recorded transfer from the known controller with a compatible onboard interval, or content corroboration across several independent anchors with no competing candidate. Time labels, directory names, largest/newest file and matching parameter hash alone are insufficient. Maintain reasons, rejected alternatives, algorithm version and a user-override audit trail.

Association state	Meaning	Product behavior
Verified	Required identity/interval evidence and corroboration pass the selected rule, with no unresolved contradiction or indistinguishable candidate.	Attach automatically and display the evidence summary. Verified means verified by that documented rule, not a universal proof of device identity.
Provisional	Likely match with an explicit weak/missing dimension, such as no UID or uncertain clock and insufficient corroboration.	Retain for review/basic exploration; withhold comparisons that depend on the missing evidence. Do not silently promote by elapsed time.
Ambiguous or Rejected	Multiple plausible candidates or contradictory identity/content.	Ask only for the specific unresolved choice or additional evidence; explain the distinguishing fields. Preserve rejected files outside confirmed bundles.
Missing or Unrecoverable	Expected source is absent, not yet transferred, never recorded or confirmed lost.	Show its consequence and exact recovery route; never substitute another flight's log.

While disarmed before flight, store an available onboard log inventory. After disarm, refresh and examine new/changed candidates. Directory timestamps can be wrong, including the earlier 2028 problem; raw source dates are retained but cannot overwrite an existing flight. Use internal monotonic/boot/event anchors and UTC-quality checks. Never treat the maximum log ID as permanently

unique: counters/files can roll over or be reused after erase/reformat. Include aircraft, inventory generation and content evidence.

A logger may keep a file open after disarm depending on settings. Stable size alone does not prove closure. Use known logger lifecycle, repeated inventory/content checks and firmware-supported closure evidence; otherwise mark the download as a provisional prefix and reconcile later. Do not silently change logging parameters or reboot the aircraft to close a file. Desired per-flight log rotation is a profile recommendation to verify separately.

After disarm, choose the fastest validated route: stable local GCS download, native file/SD access, or read-only MAVLink log transfer/MAVFTP. Show measured throughput, estimated remaining time and a profile-configured SD suggestion threshold. Capability/path differences remain version-specific. Resume by verified byte ranges/source identity; detect a growing/replaced remote file and invalidate unsafe checkpoints. Completion requires transfer integrity and parse/content checks, not simply reaching an estimated size.

For SD fallback, create a RetrievalRequest tied to the session: exact controller identity; observed remote path or protocol log ID; known mapping between ID and SD path; candidate size/stability; timestamp with quality label; boot/arming intervals; required bytes/segments; and matching rationale. Display a path/filename only if actually observed or derived by a validated firmware adapter. A protocol log ID is not automatically the SD filename.

If the mapping is known, say which exact file to copy and show size plus flight identity. If unknown, say so and offer Select the SD log folder; scan its candidates automatically and attach the matching content. The user should not need to guess which date-named file belongs to the flight. Reject a wrong import with the identity/interval mismatch and retain the outstanding request. Duplicate import is idempotent. A later correct file resolves the request and recomputes affected findings without erasing prior report history.

Build a per-flight bundle manifest referencing pre/post parameters, parameter timeline, app recorder, MP/QGC artifacts, BIN selections, RF captures, host/network/video diagnostics and hardware/test notes. Each entry records hash, bytes, origin, acquisition times, coverage, association rule/state and a recovery action. Export a portable flight bundle for support without requiring the original folder layout; redact credentials and offer location redaction separately from the local originals.

33 Mission Planner And Qgroundcontrol Integration

Detect installed/running GCS applications through allowed process/executable/version information, native settings and approved log directories. Corroborate the active GCS using socket ownership, observed connection and newly written logs. A running process or a MAV_TYPE_GCS heartbeat alone does not prove the application controlled this flight. Store process start/version/path and integration-profile revision; distinguish detected, connected, controlling and logging states.

With exactly one compatible active GCS, select its enrolled adapter automatically. With both MP and QGC active, do not choose whichever launched last. Identify the intended control connection and ask once if it remains ambiguous; remember the choice for that topology. An in-flight GCS switch is a recorded session event and may add a second GCS artifact without creating a new flight. Never terminate the other program or rewrite its settings while flying.

The preferred zero-touch telemetry architecture is a separately supervised, bounded local connection service with explicit fan-out and read-only app requests. Where possible preserve a direct GCS control connection and obtain duplication through a supported independent endpoint. Where a single UDP/serial owner is unavoidable, an explicitly commissioned router owns it and passes the unchanged GCS traffic while feeding the recorder and narrowly filtered app request channel. The recording UI/analysis process

must not be the transport's lifetime owner.

This routing service is a critical dependency requiring bench validation before flight use. It must not fabricate, cache/replay or continue GCS heartbeats/manual control after the GCS dies, nor hide RC/GCS failsafe loss by impersonation. Keep system/component IDs and signing behavior correct, prevent routing loops and duplicate commands, and never forward arbitrary renderer-generated MAVLink. A recorder backlog or UI crash must not block the live transport. Test router failure and provide a documented direct-GCS fallback that is established on the ground; do not promise seamless failover over an exclusive serial port.

Read-only collection still sends messages. Allow specific identity/parameter/log-read operations with exact target checks and bounded rates; inspect MAVFTP opcodes, not only FILE_TRANSFER_PROTOCOL message IDs. Deny upload/write/delete/reset/format/parameter-set/arm/mode/mission/control operations from the app. Do not use FTP ResetSessions indiscriminately because it may interrupt another client's transfer. Coordinate one bulk transaction owner with the GCS. Respect signed links and credential boundaries; never disable signing to make collection work.

Integration	Automatic behavior	Connection limitation
Mission Planner	Discover the actual configured log directory, watch stable TLOGs and related diagnostic logs, record loaded GStreamer/config versions, and collect relevant file ranges after flight.	Forwarding behavior depends on settings/version. A remote-write/control option is not automatically safe; use an explicit topology and app-side read-only filter, not blanket remote control enablement [R37].
QGroundControl	Discover its version-specific save/logging settings and authorized data directory, watch saved TLOGs and appropriate application logs, and inspect its own video settings.	Documented MAVLink forwarding is one-way to the destination; it cannot carry the app's parameter/log requests back [R36]. Use an approved bidirectional router/independent path or mark automatic requests unavailable. PX4 log streaming/upload settings are not ArduPilot BIN capture.
App-owned recorder	Start when the feed is ready; persist compatible timestamped MAVLink plus richer origin/direction/clock metadata, and segment by session without losing originals.	A receive-only forwarded feed may omit GCS-sent messages. Do not label it a full duplex GCS TLOG or claim it contains every command. Preserve native GCS TLOGs as complementary artifacts.

Default folders are discovery hints, not a hardcoded contract. Handle portable installs, moved Documents directories, per-user profiles, sandbox/package permissions and custom paths. Ask once for an inaccessible folder. QGC save prompts/settings and GCS file handles can delay finalization until disconnect or exit; do not disconnect a control link to obtain a file. Display Awaiting GCS file close while using the app's recorder. Treat a copied open file as a partial prefix until reconciled.

For setup errors, show the failed capability, current evidence and a single relevant action: enable the supported log-save setting, grant a folder, choose the intended GCS, repair the endpoint, install/repair a verified decoder, or select an approved router profile. Back up GCS configuration and validate a version-specific patch before offering an automatic edit; otherwise provide precise guided steps. Verify actual saved files and fresh packets after setup, not only checkbox state.

34 Configuration-To-Performance Insights

Every finding and metric selection links to a ParameterTimeline segment and evidence manifest revision. Show the actual configuration during that selected interval alongside the data. Do not compare against a stale profile default or whichever .param the operator happened to upload last. An unknown effective-change boundary blocks attribution across that boundary but does not block descriptive metrics within supported scopes.

MetricResult and Finding require flight_record_id, armed_interval_id, capture_session_id, evidence_manifest_revision, effective_configuration_interval_ids and configuration_eligibility with reasons. If configuration evidence is unavailable, retain the explicit unknown state and empty interval references; permit descriptive measurements that do not need it and withhold unsupported parameter attribution. A metric spanning different effective configurations must be partitioned or explicitly marked mixed, not silently assigned one snapshot.

Provide a Changes & effects view: previous comparable flight, current flight, firmware/hardware identity, changed parameter groups, relevant metrics, absolute/relative differences, coverage/uncertainty and competing explanations. Clicking a parameter highlights affected signal workspaces and its firmware-specific meaning, units, reboot behavior and dependencies. Clicking a performance regression opens both aligned, phase-matched chart selections.

Parameter group	Relevant outcomes	Interpretation gate
PID / attitude / rate	Rate tracking error, response delay/overshoot, oscillatory content, actuator authority/saturation, PID term behavior and noise.	Representative excitation and comparable mode/phase; resolve vibration/filter/estimator confounders first. No gain prescription from RMSE or power alone.
Gyro / notch / filter	Pre/post gyro spectra, tracked motor harmonics, modeled attenuation/phase delay, tracking and CPU cost.	Correct sample timing, pre/post provenance, actual RPM source and firmware algorithm. Modeled effects are not measured flight improvement.
Motor / hover / payload	Hover demand, output distribution, RPM/load balance, authority, climb response and thermal behavior across actual all-up mass/CG.	Store battery/payload/prop changes; distinguish normalized control demand from physical thrust. Do not assume a displayed throttle percentage is motor duty cycle or thrust percentage.
Battery / energy	W, Wh, hover W/kg when actual mass is known, distance-normalized energy for an appropriate phase, voltage sag and consistent capacity use.	Calibration/sign/monitor identity and matched conditions; changing a current sensor scale can change reported energy without changing real consumption.
Navigation / estimator / failsafe	Tracking, drift/heading consistency, innovations, mode transitions and observed trigger/response behavior.	Firmware-specific semantics and navigation quality; a changed failsafe threshold is configuration evidence, not proof the failsafe works.
Logging / network / video	Sample completeness, transfer/decoder statistics, freshness, drop/freeze intervals and host load.	A new logging rate can change observed extrema or spectra. A different radio/camera/GCS/network profile can explain apparent control/video changes independently of ArduPilot tuning.

Track multiple throttle quantities explicitly: pilot input, collective/control demand, learned hover reference and per-motor output, with actual field, scale and units. Display percent only when the source range and normalization are known. Preserve raw DShot/PWM/output units and do not average incompatible quantities into one throttle number.

Require a comparison eligibility checklist computed from actual data: aircraft/hardware revision; firmware; effective parameter interval; mass/payload/CG; battery ID, state, temperature and calibration; prop/motor changes; mode and motion phase; wind evidence; sensor/logging coverage. Missing conditions are visible. Nearby weather/API data is environmental context, not measured wind at the aircraft. Use repeated trials and uncertainty at the flight/block level, not millions of correlated samples as independent trials.

Recommendations progress from observed difference to plausible explanation to proposed experiment. Example wording may say Lower hover power was observed after the filter change, but payload and wind also changed; retest under matched conditions. It must not say The filter caused the endurance gain without appropriate evidence. Keep successful response/thermal/authority constraints beside energy goals. Parameter suggestions need exact firmware applicability, prerequisites, evidence, trade-offs and a rollback snapshot; when insufficient, recommend the next measurement instead of a guessed value.

Support variable-payload cohorts and regression history across accepted tunes. Store payload/mass as first-class fields and compare light/heavy configurations within validated envelopes. Preserve pre-FACt, FACt-derived and later validated configurations; never relabel an unvalidated inherited notch tune as accepted merely because its source aircraft was similar.

35 Implementation Contract Extensions

Add native NetworkInspector and RepairBroker adapters, a SessionOrchestrator, ParameterCaptureService, EvidenceMatcher, GCS adapters, a separately supervised ConnectionService, and a configuration-performance index. Keep them separate from numerical diagnostics. The renderer receives structured findings/plans and cannot execute arbitrary commands. A network repair has a different permission boundary from a data collector and from aircraft tuning proposals.

New persisted entities: NetworkProfileVersion, NetworkObservation, RepairPlan/RepairTransaction, CaptureSession, ArmedInterval, GcsObservation, ParameterCapture, ParameterObservation, EffectiveConfigurationInterval, ArtifactAssociation, RetrievalRequest and RecorderCoverage. Each records version, origin and immutable evidence references. Store IP addresses as typed addresses/prefixes; stable interface IDs separately from user-editable names; UTC/monotonic times separately; 64-bit JSON times as decimal strings.

Extend IPC with network.inspect, network.plan_repair, network.apply_repair, network.rollback, session.prepare_and_watch, session.status, parameters.capture, parameters.reconcile, associations.evaluate, associations.review and retrieval.resolve. Every mutation requires an operation ID, target profile version and applicability fingerprint; repair apply additionally requires the reviewed plan token and current preconditions. Events include source freshness, flight state, parameter readiness, association status, missing artifact, repair progress/outcome and disk pressure.

The bundled JSON contract is normative for required fields/state names/rule IDs, with this text explaining behavior. Codex should turn these design contracts into versioned runtime schemas, not assume the handoff JSON is already a working API or executable test suite. Treat all acceptance cases here as required tests not yet executed.

Revise the implementation order so automatic capture/identity is foundational. Stage 0 proves packaging plus transport/process isolation and a network inventory spike. Stage 1 implements

append-only recording, aircraft identity, pre/post parameter capture, session state and evidence manifests using real fixtures. Stage 2 implements MP/QGC adapters, verified file association, BIN retrieval/SD requests and crash/reconnect behavior. Stage 3 integrates the original health, vibration, filter, PID and energy analysis with effective configuration intervals. Stage 4 qualifies OS repair adapters and cross-platform stream diagnostics; stage 5 qualifies installers, operational behavior, regressions and one-click complete workflows. Native repair is not required to run the early read-only analyzer.

Add a persistent readiness strip with aircraft, network, telemetry, parameters, radio recorder, disk and GCS states. Each badge opens an actionable finding. Network & devices shows the selected Internet/drone interfaces, route/source decisions, receiving apps and transport/decode status. Keep technical details collapsed by default. A repair card states the concrete change and interruption before Apply; a collection card shows the exact missing artifact and fastest recovery. Avoid repeated success dialogs and repeated confirmation for already configured read-only collection.

Commission once with bench evidence: both GCS versions; installed firmware/UID availability; verified topology and source IPs; sample SkyPulse responses; actual BIN/TLOG/log paths and lifecycle; OS permission model; decoder runtime; link bandwidth; rearm/reboot/log rollover cases; and documented control-loss behavior with the recorder running. These are implementation validation inputs, not blockers to completing this design.

36 Required Scenarios And Release Gates

Acceptance	Required result
Simultaneous Internet and drone network	Wi-Fi Internet remains working while Ethernet 10.0.0.5/24 receives from 10.0.0.2. No added drone default gateway, bridge or NAT. Same-NIC DHCP and VPN overlap cases produce a tested plan or guided limitation, not a damaging generic fix.
Network repair lifecycle	Preview, declined elevation, managed-policy block, duplicate request, stale plan, concurrent settings change, UI crash and failed verification all preserve an audit trail. Rollback changes only owned objects. Armed/unknown state blocks disruptive repair.
UDP and video diagnosis	Distinguish firewall suspicion from a proven block; packet arrival from decode; wrong H.264/H.265 from missing plugins; port conflict from bad route; RTSP authentication from ICMP silence. MP wrong-runtime and QGC independent-runtime cases are covered.
GCS coexistence and transport	MP and QGC both running do not cause arbitrary owner selection or double bind. QGC one-way forwarding cannot falsely report readback/download capability. App UI/FFT/recorder crashes do not spoof GCS heartbeat or commands; loss of controlling GCS remains visible to the aircraft under the actual failsafe configuration.
Automatic session boundaries	Capture begins before arm; normal arm/disarm produces one interval; rapid rearm remains distinct; armed bench test is not a flight; app starts midflight as partial; heartbeat loss never means landed; replay cannot open a live flight; reboot and two same-sysid aircraft do not merge.
Parameter capture correctness	Missing indexes, changing counts, duplicate conflicts, wrong encoding, FTP fallback, live value changes during download, absent update messages, reboot-required settings and stale cache all receive correct completeness/freshness/effective-time states. No count-complete download is claimed atomic.
File association and recovery	Wrong 2028 timestamps, same filenames with different bytes, reused log IDs, overlapping flights, one BIN with many arm cycles, one flight with multiple files, missing UID and contradictory parameter evidence cannot silently misassociate. Known manual SD requests name the observed file; unknown names trigger content scanning, not a guessed filename.
Post-flight collection	Open/growing BIN/TLOG stays provisional; GCS holds a file until exit without forced disconnect; slow transfer offers measured ETA and SD option; interruption resumes only after identity validation; rearming pauses transfer; a late correct import resolves only the appropriate RetrievalRequest.
Short-lived and incomplete data	RF polling starts preflight, retention overruns and auth/endpoint errors are recorded, unreachable air radio is explicit, sleep/crash/disk-pressure gaps remain visible. Optional video cannot starve the essential recorder or control transport.
Performance comparison	Relevant parameter changes split eligible intervals, unknown boundaries withhold attribution, mass/battery/wind/logging-rate changes flag confounding, current-scale edits cannot masquerade as energy savings, and no high-current-only PID recommendation appears. Pre-FACt/FACt baselines retain original bytes and alias history.
Cross-platform qualification	Publish a tested OS/GCS/version/capability matrix. Permissions, install/signing, access to local-network/packet observations and rollback behavior are exercised on actual supported OS installations. Mocks are used for fault injection, not claimed as hardware validation.
User-facing completion	An ordinary commissioned flight runs Prepare & watch -> automatic capture -> postflight collection -> evidence-linked report without manual file pairing. A degraded flight shows one precise recovery task per gap and preserves useful results; it never reports complete coverage merely to avoid asking a question.

Primary references

Original research: 3 October 2026. Revision-2 additions: 12 October 2026. Firmware and application sources must be pinned to the implemented versions; live documentation is not a compatibility guarantee.

[R01] Tauri 2: external binaries / sidecars

https://v2.tauri.app/develop/sidecar/

Native shell and packaged engine capability; architecture choice is our recommendation.

[R02] DuckDB: Parquet overview

https://duckdb.org/docs/stable/data/parquet/overview

Column and time-window query design.

[R03] Apache ECharts

https://echarts.apache.org/en/index.html

Chart library capabilities; app performance must be measured.

[R04] ArduPilot: measuring vibration

https://ardupilot.org/copter/docs/common-measuring-vibration.html

Vibration bands and clipping interpretation.

[R05] ArduPilot: raw IMU logging

https://ardupilot.org/copter/docs/common-raw-imu-logging.html

Raw gyro/filter evidence requirements.

[R06] ArduPilot: harmonic notch filtering

https://ardupilot.org/copter/docs/common-imu-notch-filtering.html

Filter modes and trade-offs; exact masks remain firmware-specific.

[R07] ArduPilot WebTools repository / FilterReview

https://github.com/ArduPilot/WebTools

Reference implementations and license review; pin commits before reuse.

[R08] ArduPilot PIDReview README

https://github.com/ArduPilot/WebTools/blob/main/PIDReview/Readme.md

PID/response analysis prerequisites and rate-only limitations.

[R09] ArduPilot: ESC telemetry

https://ardupilot.org/copter/docs/common-esc-telemetry.html

Telemetry capabilities differ by firmware and protocol.

[R10] DigiQ SkyPulse manufacturer manual (version-sensitive URL)

https://digiq-rf.com/datasheets/DigiQ_RF_SkyPulse_User_Guide_Manual.pdf

Rechecked 12 October: destination learning, rolling histories and API path/reboot caveats. Pin PDF/version and installed firmware before implementing.

[R11] MAVLink: log transfer protocol

https://mavlink.io/en/services/log.html

Controller log retrieval protocol.

[R12] ArduPilot: MAVLink log transfer

https://ardupilot.org/dev/docs/mavlink-log-download.html

Version/capability differences for retrieval.

[R13] MAVLink common messages: SYSTEM_TIME

https://mavlink.io/en/messages/common.html#SYSTEM_TIME

Unix and boot clock fields; validity must be checked.

[R14] ArduPilot: onboard log message catalog

https://ardupilot.org/copter/docs/logmessages.html

Representative field discovery; use matching source/FMT for exact build.

[R15] ArduPilot: log download and analysis

https://ardupilot.org/copter/docs/common-downloading-and-analyzing-data-logs-in-mission-planner.html

Logging configuration and dropped-message diagnostics.

[R16] ArduPilot pymavlink DFReader

https://github.com/ArduPilot/pymavlink/blob/master/DFReader.py

Decoding/scaling/time behavior; pin parser revision.

[R17] ArduPilot: diagnosing problems using logs

https://ardupilot.org/copter/docs/common-diagnosing-problems-using-logs.html

Health review context.

[R18] ArduPilot: tuning process

https://ardupilot.org/copter/docs/tuning-process-instructions.html

Mechanical/filter/control tuning order.

[R19] ArduPilot: SIYI camera/gimbal integration

https://ardupilot.org/copter/docs/common-siyi-zr10-gimbal.html

Model/firmware-dependent integration and media retrieval.

[R20] ArduPilot Copter-4.7.1 source tree

https://github.com/ArduPilot/ardupilot/tree/Copter-4.7.1

Version-specific implementation authority when the aircraft matches this tag.

[R21] Copter-4.7.1: logger structures

https://raw.githubusercontent.com/ArduPilot/ardupilot/Copter-4.7.1/libraries/AP_Logger/LogStructure.h

Schema and event/command fields.

[R22] Copter-4.7.1: IMU structures

https://raw.githubusercontent.com/ArduPilot/ardupilot/Copter-4.7.1/libraries/AP_InertialSensor/LogStructure.h

Modern VIBE, raw and batch fields.

[R23] Copter-4.7.1: EKF outputs

https://raw.githubusercontent.com/ArduPilot/ardupilot/Copter-4.7.1/libraries/AP_NavEKF3/AP_NavEKF3_Outputs.cpp

Test metric transformations and meaning.

[R24] Copter-4.7.1: EKF structures

https://raw.githubusercontent.com/ArduPilot/ardupilot/Copter-4.7.1/libraries/AP_NavEKF3/LogStructure.h

Lane/source/innovation metadata.

[R25] Copter-4.7.1: battery logging

https://raw.githubusercontent.com/ArduPilot/ardupilot/Copter-4.7.1/libraries/AP_BattMonitor/AP_BattMonitor_Logging.cpp

Consumption units and sentinel/encoding behavior.

[R26] Copter-4.7.1: ESC structures

https://raw.githubusercontent.com/ArduPilot/ardupilot/Copter-4.7.1/libraries/AP_ESC_Telem/LogStructure.h

ESC schemas; verify actual installed firmware.

[R27] Copter-4.7.1: scheduler

https://raw.githubusercontent.com/ArduPilot/ardupilot/Copter-4.7.1/libraries/AP_Scheduler/AP_Scheduler.cpp

Scheduler metrics and load encoding.

[R28] Copter-4.7.1: attitude/position structures

https://raw.githubusercontent.com/ArduPilot/ardupilot/Copter-4.7.1/libraries/AC_AttitudeControl/LogStructure.h

Modern PSC, RATE and control units.

[R29] Copter-4.7.1: mission structures

https://raw.githubusercontent.com/ArduPilot/ardupilot/Copter-4.7.1/libraries/AP_Mission/LogStructure.h

Mission logging semantics.

[R30] ArduPilot: MAVFTP

https://ardupilot.org/dev/docs/mavlink-mavftp.html

Optional transport and version-specific paths.

[R31] EdgeTX: special functions / SD Logs

https://manual.edgetx.org/color-radios/model-settings/special-functions

Optional radio-side CSV when SD logging is enabled; available values depend on inputs.

[R32] ArduPilot batch sampler implementation

https://github.com/ArduPilot/ardupilot/blob/master/libraries/AP_InertialSensor/BatchSampler.cpp

Batch scale/encoding reference; pin matching revision before implementation.

[R33] Copter-4.7.1: EKF failsafe logic

https://raw.githubusercontent.com/ArduPilot/ardupilot/Copter-4.7.1/ArduCopter/ekf_check.cpp

Configured threshold and filtered/state-dependent failsafe behavior.

[R34] MAVLink parameter protocol

https://mavlink.io/en/services/parameter.html

Completeness, type handling and limits of passive cache synchronization.

[R35] MAVLink packed parameter FTP

https://mavlink.io/en/services/parameter_ftp.html

Non-atomic live values, format, EOF and fallback requirements; validate installed implementation.

[R36] QGroundControl telemetry settings

https://docs.qgroundcontrol.com/master/en/qgc-user-guide/settings_view/telemetry.html

Documented one-way forwarding. Master documentation is discovery guidance; pin supported release.

[R37] Mission Planner forwarding tools

https://ardupilot.org/planner/docs/common-mp-tools.html

Local forwarding and separate remote-write/control option.

[R38] Microsoft New-NetIPAddress

https://learn.microsoft.com/en-us/powershell/module/nettcpip/new-netipaddress

DHCP-disable behavior and address configuration caveats.

[R39] QGroundControl video settings

https://docs.qgroundcontrol.com/master/en/qgc-user-guide/settings_view/video.html

Native QGC video configuration; validate installed release/runtime.

[R40] Apple firewall settings

https://support.apple.com/guide/mac-help/change-firewall-settings-on-mac-mh11783/mac

Application/service firewall behavior and policy differences.

[R41] NetworkManager D-Bus API

https://networkmanager.dev/docs/api/latest/gdbus-org.freedesktop.NetworkManager.html

Checkpoint and rollback support; target installed manager/version.

[R42] Microsoft firewall rules

https://learn.microsoft.com/en-us/powershell/module/netsecurity/new-netfirewallrule

Typed rule scope; effective policy still requires verification.

[R43] Microsoft UDP socket reuse

https://learn.microsoft.com/en-us/windows/win32/winsock/using-so-reuseaddr-and-so-exclusiveaddruse

Socket sharing is not a reliable unicast duplication mechanism.

[R44] GStreamer UDP source

https://gstreamer.freedesktop.org/documentation/udp/udpsrc.html

Receive-buffer semantics and source properties.

[R45] GStreamer D3D11 H.265 decoder

https://gstreamer.freedesktop.org/documentation/d3d11/d3d11h265dec.html

Windows decoder capability; installed GPU/runtime must be validated.

[R46] Apple Local Network privacy

https://developer.apple.com/documentation/technotes/tn3179-understanding-local-network-privacy

Separate OS permission boundary from firewall/network configuration.

[R47] QGroundControl general settings

https://docs.qgroundcontrol.com/master/en/qgc-user-guide/settings_view/general.html

Version-specific application save paths/settings.

[R48] Mission Planner telemetry logs

https://ardupilot.org/planner/docs/mission-planner-telemetry-logs.html

Native telemetry capture and log lifecycle discovery.

[R49] NetworkManager IPv4 profile

https://networkmanager.dev/docs/api/latest/settings-ipv4.html

Addresses, routes and never-default configuration.

[R50] firewalld rich language

https://firewalld.org/documentation/man-pages/firewalld.richlanguage.html

Zone/source/port rules; active policy must be identified.

[R51] Copter-4.7.1 parameter implementation

https://github.com/ArduPilot/ardupilot/blob/Copter-4.7.1/libraries/GCS_MAVLink/GCS_Param.cpp

Tagged-source check: parameter write/readback/logging behavior is not a reliable universal broadcast contract.

[R52] Copter-4.7.1 parameter logging

https://github.com/ArduPilot/ardupilot/blob/Copter-4.7.1/libraries/AP_Logger/LogFile.cpp

PARM observation timestamp semantics; pin matching firmware source.

[R53] Apple SystemConfiguration

https://developer.apple.com/documentation/systemconfiguration

Native network service inspection/configuration APIs.